Week3Deliverable

Refined Questions and Analysis Choices

Central Question:

What is the relationship between temporal / land cover factors and fast fire events (high mean growth rate per event)?

#Supporting Questions: What is the relationship between temporal /land cover factors and the longest fire events (duration)? Is there a relationship between land cover and maximum, mean, minimum daily fire growth per event?

Observations and Variables

There are a total of 4329 observations (fire events) available to use

Variable: ig_month Description: The month of the earliest date contained in the event

Variable: ig_year Description: The year of the earliest date contained in the event.

Variable: event_dur Description: Duration of the event in days

Variable: mx_grw_km2 Description: Maximum daily fire growth per event in square kilometers

Variable: mn_grw_km2 Description: Minimum daily fire growth per event in square kilometers

Variable: mu_grw_km2 Description: Mean daily fire growth per event in square kilometers

Variable: lc_name Description: Character string of the landcover type from the year before the fire.

Intended Comparison / Unresolved Data Limitations

An intended comparison is the central question and first supporting question. Addressing whether the temporal / land factors are contributing more to the mean growth rate of the fire events or the duration of the fire event.

There are no unresolved data limitations, since there are no gaps for the intended variables.

Purposeful Explanatory Evidence

```
#Loading packages and setting working directory
library(tidyverse)
library(here)
here()
```

```
## [1] "/home/guest/EDE_II/BelusWarrenKeeleyTauber"
```

```
#install.packages("agricolae")
library(agricolae)
library(lubridate)
library(ggthemes)
library(ggplot2)
library(dplyr)
library(gt)

california_events_processed <- read_csv(here("Data/Processed/california_events_processed.csv"))

# Changing lc_name variable from character to factor
california_events_processed$lc_name <- as.factor(california_events_processed$lc_name)

# Changing eco_name variable from character to factor
california_events_processed$eco_name <- as.factor(california_events_processed$eco_name)

# changing month to factor format
california_events_processed$ig_month <- as.factor(california_events_processed$ig_month)

#changing year to factor format
california_events_processed$ig_year <- as.factor(california_events_processed$ig_year)

#creating the theme
theme <- theme_bw() +
  #making the title bold, 13 font, and centered
  theme(plot.title = element_text(face = "bold", size = 10, hjust = 0.5),
        #making my axis titles bold and 10 font
          axis.title = element_text(face = "bold", size = 10),
        #positioning my legend on the right side of the plot
        legend.position = "right"
  )

#setting the theme
theme_set(theme)

#Establishing the color palette
canva_pal("Birds and berries")
```

```
## function (n)
## {
##     n_values <- length(values)
##     if (n > n_values) {
##         cli::cli_warn("This manual palette can handle a maximum of {n_values} values. You have suppli
##     }
##     unname(values[seq_len(n)])
## }
## <bytecode: 0x5acc4d0f7188>
## <environment: 0x5acc4d0fa420>
```

What is the relationship between time of year and which land cover types see the fastest fires?

Response variable: mean growth of fire per day; Predictors: land cover and month

```
#We chose to use two-way ANOVA to explore if the predictors of land cover and month may cause
#a variation in the response variable, mean growth of fire per day.

# create new data frame of just question variables
mean_growth <- california_events_processed %>%
  select(lc_name, ig_month, mu_grw_km2)

#viewing the data frame of variables and commenting it out after
#view(mean_growth)

# Format as aov
mean_growth_aov2 <- aov(data = mean_growth, mu_grw_km2 ~ ig_month + lc_name)
summary(mean_growth_aov2)
```

```
##               Df Sum Sq Mean Sq F value   Pr(>F)
## ig_month      11  15911  1446.4   8.885 7.88e-16 ***
## lc_name       13  23239  1787.6  10.981  < 2e-16 ***
## Residuals   4304 700685   162.8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Format as lm
mean_growth_anova_2way2 <- lm(data = mean_growth, mu_grw_km2 ~ ig_month + lc_name)
summary(mean_growth_anova_2way2)
```

```
##
## Call:
## lm(formula = mu_grw_km2 ~ ig_month + lc_name, data = mean_growth)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.377  -3.531  -1.603   0.390 244.029
##
## Coefficients:
##                                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                               -1.7439     1.5703  -1.111 0.266818
## ig_month2                                  0.5995     1.4335   0.418 0.675798
## ig_month3                                  0.9050     1.4934   0.606 0.544560
## ig_month4                                  0.4283     1.4261   0.300 0.763929
## ig_month5                                 -0.9559     1.2918  -0.740 0.459327
## ig_month6                                 -0.2234     1.2541  -0.178 0.858651
## ig_month7                                  2.7056     1.2557   2.155 0.031237
## ig_month8                                  4.0380     1.2619   3.200 0.001385
## ig_month9                                  1.8168     1.2254   1.483 0.138247
## ig_month10                                 2.9066     1.2433   2.338 0.019439
## ig_month11                                 0.7028     1.2461   0.564 0.572808
## ig_month12                                 1.2472     1.6310   0.765 0.444521
```

```
## lc_nameClosed Shrublands                    10.2979     1.6233   6.344 2.47e-10
## lc_nameCropland/Natural Vegetation Mosaics   2.6451     4.2128   0.628 0.530116
## lc_nameCroplands                             1.1229     1.1889   0.945 0.344956
## lc_nameEvergreen Broadleaf Forests           8.8021     4.4030   1.999 0.045657
## lc_nameEvergreen Needleleaf Forests          6.1481     1.3488   4.558 5.30e-06
## lc_nameGrasslands                            5.0559     1.1715   4.316 1.63e-05
## lc_nameOpen Shrublands                       9.4173     1.6524   5.699 1.28e-08
## lc_namePermanent Snow and Ice               -0.2967     5.8160  -0.051 0.959320
## lc_namePermanent Wetlands                    0.2087     4.9501   0.042 0.966372
## lc_nameSavannas                              4.2843     1.3982   3.064 0.002197
## lc_nameUrban and Built-up Lands              1.4826     3.8642   0.384 0.701241
## lc_nameWater Bodies                          0.3313     7.4550   0.044 0.964551
## lc_nameWoody Savannas                        4.6332     1.3937   3.324 0.000894
##
## (Intercept)
## ig_month2
## ig_month3
## ig_month4
## ig_month5
## ig_month6
## ig_month7                                  *
## ig_month8                                  **
## ig_month9
## ig_month10                                 *
## ig_month11
## ig_month12
## lc_nameClosed Shrublands                   ***
## lc_nameCropland/Natural Vegetation Mosaics
## lc_nameCroplands
## lc_nameEvergreen Broadleaf Forests         *
## lc_nameEvergreen Needleleaf Forests        ***
## lc_nameGrasslands                          ***
## lc_nameOpen Shrublands                     ***
## lc_namePermanent Snow and Ice
## lc_namePermanent Wetlands
## lc_nameSavannas                            **
## lc_nameUrban and Built-up Lands
## lc_nameWater Bodies
## lc_nameWoody Savannas                      ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.76 on 4304 degrees of freedom
## Multiple R-squared:  0.05292,    Adjusted R-squared:  0.04764
## F-statistic: 10.02 on 24 and 4304 DF,  p-value: < 2.2e-16
```

```r
# Run a post-hoc test for pairwise differences
TukeyHSD(mean_growth_aov2)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = mu_grw_km2 ~ ig_month + lc_name, data = mean_growth)
##
```

```
## $ig_month
##               diff        lwr         upr     p adj
## 2-1   -0.18301466 -4.8547420  4.48871270 1.0000000
## 3-1   -0.20900982 -5.0739895  4.65596984 1.0000000
## 4-1    0.45304344 -4.1906500  5.09673684 1.0000000
## 5-1    0.33726472 -3.8256864  4.50021583 1.0000000
## 6-1    1.40337019 -2.6393112  5.44605161 0.9931178
## 7-1    4.29617553  0.2590180  8.33333311 0.0254409
## 8-1    5.80526225  1.7372130  9.87331147 0.0001989
## 9-1    2.10512692 -1.8911208  6.10137467 0.8581289
## 10-1   3.01713082 -1.0370335  7.07129512 0.3834289
## 11-1   0.28109849 -3.7816443  4.34384129 1.0000000
## 12-1   1.35115632 -3.9773546  6.67966724 0.9996023
## 3-2   -0.02599515 -4.4434624  4.39147211 1.0000000
## 4-2    0.63605811 -3.5364565  4.80857274 0.9999977
## 5-2    0.52027938 -3.1096176  4.15017635 0.9999988
## 6-2    1.58638486 -1.9049276  5.07769736 0.9446962
## 7-2    4.47919020  0.9942754  7.96410502 0.0016073
## 8-2    5.98827691  2.4676215  9.50893232 0.0000019
## 9-2    2.28814158 -1.1492973  5.72558047 0.5669019
## 10-2   3.20014548 -0.3044570  6.70474792 0.1129598
## 11-2   0.46411315 -3.0504094  3.97863575 0.9999995
## 12-2   1.53417098 -3.3891428  6.45748478 0.9973091
## 4-3    0.66205326 -3.7257559  5.04986238 0.9999979
## 5-3    0.54627454 -3.3291793  4.42172838 0.9999990
## 6-3    1.61238001 -2.1335850  5.35834498 0.9623465
## 7-3    4.50518535  0.7651824  8.24518827 0.0047414
## 8-3    6.01427206  2.2409440  9.78760018 0.0000127
## 9-3    2.31413674 -1.3816685  6.00994200 0.6605292
## 10-3   3.22614063 -0.5322139  6.98449516 0.1772693
## 11-3   0.49010831 -3.2774983  4.25771492 0.9999996
## 12-3   1.56016613 -3.5468887  6.66722098 0.9977472
## 5-4   -0.11577872 -3.7095238  3.47796637 1.0000000
## 6-4    0.95032675 -2.5033836  4.40403705 0.9991374
## 7-4    3.84313209  0.3958892  7.29037493 0.0143088
## 8-4    5.35221880  1.8688488  8.83558877 0.0000339
## 9-4    1.65208347 -1.7471573  5.05132426 0.9133200
## 10-4   2.56408737 -0.9030570  6.03123174 0.3937134
## 11-4  -0.17194495 -3.6491163  3.30522644 1.0000000
## 12-4   0.89811287 -3.9986075  5.79483326 0.9999842
## 6-5    1.06610547 -1.7077706  3.83998154 0.9841211
## 7-5    3.95891081  1.1930914  6.72473018 0.0001859
## 8-5    5.46799753  2.6572788  8.27871629 0.0000001
## 9-5    1.76786220 -0.9378931  4.47361745 0.5963503
## 10-5   2.67986610 -0.1107187  5.47045092 0.0736228
## 11-5  -0.05616623 -2.8591993  2.74686686 1.0000000
## 12-5   1.01389160 -3.4295467  5.45732992 0.9998575
## 7-6    2.89280534  0.3115522  5.47405849 0.0133577
## 8-6    4.40189205  1.7725859  7.03119824 0.0000031
## 9-6    0.70175672 -1.8150314  3.21854486 0.9990239
## 10-6   1.61376062 -0.9940114  4.22153261 0.6773683
## 11-6  -1.12227170 -3.7433603  1.49881691 0.9637657
## 12-6  -0.05221388 -4.3831786  4.27875082 1.0000000
## 8-7    1.50908672 -1.1117184  4.12989181 0.7701817
```

```
## 9-7   -2.19104861 -4.6989543  0.31685706 0.1571295
## 10-7  -1.27904472 -3.8782452  1.32015575 0.9061654
## 11-7  -4.01507704 -6.6276378 -1.40251625 0.0000337
## 12-7  -2.94501922 -7.2708282  1.38078979 0.5305000
## 9-8   -3.70013533 -6.2574726 -1.14279803 0.0001458
## 10-8  -2.78813143 -5.4350591 -0.14120373 0.0286293
## 11-8  -5.52416376 -8.1842121 -2.86411544 0.0000000
## 12-8  -4.45410593 -8.8087594 -0.09945248 0.0395852
## 10-9   0.91200390 -1.6231880  3.44719576 0.9907222
## 11-9  -1.82402843 -4.3729161  0.72485929 0.4480491
## 12-9  -0.75397060 -5.0416248  3.53368356 0.9999897
## 11-10 -2.73603233 -5.3747973 -0.09726732 0.0341653
## 12-10 -1.66597450 -6.0076597  2.67571071 0.9843242
## 12-11  1.07005783 -3.2796389  5.41975451 0.9997029
##
## $lc_name
##                                                                        diff
## Closed Shrublands-Barren                                         10.5113510
## Cropland/Natural Vegetation Mosaics-Barren                        4.0268351
## Croplands-Barren                                                  2.1321147
## Evergreen Broadleaf Forests-Barren                                8.7342961
## Evergreen Needleleaf Forests-Barren                               6.2707278
## Grasslands-Barren                                                 5.1558302
## Open Shrublands-Barren                                            9.4636340
## Permanent Snow and Ice-Barren                                    -0.8107703
## Permanent Wetlands-Barren                                         0.1466988
## Savannas-Barren                                                   4.7172442
## Urban and Built-up Lands-Barren                                   2.2466799
## Water Bodies-Barren                                               0.9646244
## Woody Savannas-Barren                                             4.8874296
## Cropland/Natural Vegetation Mosaics-Closed Shrublands            -6.4845160
## Croplands-Closed Shrublands                                      -8.3792363
## Evergreen Broadleaf Forests-Closed Shrublands                    -1.7770550
## Evergreen Needleleaf Forests-Closed Shrublands                   -4.2406232
## Grasslands-Closed Shrublands                                     -5.3555208
## Open Shrublands-Closed Shrublands                                -1.0477171
## Permanent Snow and Ice-Closed Shrublands                        -11.3221213
## Permanent Wetlands-Closed Shrublands                            -10.3646522
## Savannas-Closed Shrublands                                       -5.7941068
## Urban and Built-up Lands-Closed Shrublands                       -8.2646711
## Water Bodies-Closed Shrublands                                   -9.5467266
## Woody Savannas-Closed Shrublands                                 -5.6239214
## Croplands-Cropland/Natural Vegetation Mosaics                    -1.8947203
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics   4.7074610
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics  2.2438928
## Grasslands-Cropland/Natural Vegetation Mosaics                    1.1289952
## Open Shrublands-Cropland/Natural Vegetation Mosaics               5.4367989
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics       -4.8376053
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics           -3.8801363
## Savannas-Cropland/Natural Vegetation Mosaics                      0.6904091
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics     -1.7801551
## Water Bodies-Cropland/Natural Vegetation Mosaics                 -3.0622107
## Woody Savannas-Cropland/Natural Vegetation Mosaics                0.8605946
## Evergreen Broadleaf Forests-Croplands                            6.6021814
```

```
## Evergreen Needleleaf Forests-Croplands                                4.1386131
## Grasslands-Croplands                                                  3.0237155
## Open Shrublands-Croplands                                             7.3315193
## Permanent Snow and Ice-Croplands                                     -2.9428850
## Permanent Wetlands-Croplands                                         -1.9854159
## Savannas-Croplands                                                    2.5851295
## Urban and Built-up Lands-Croplands                                    0.1145652
## Water Bodies-Croplands                                               -1.1674903
## Woody Savannas-Croplands                                              2.7553149
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests             -2.4635683
## Grasslands-Evergreen Broadleaf Forests                               -3.5784658
## Open Shrublands-Evergreen Broadleaf Forests                           0.7293379
## Permanent Snow and Ice-Evergreen Broadleaf Forests                   -9.5450663
## Permanent Wetlands-Evergreen Broadleaf Forests                       -8.5875973
## Savannas-Evergreen Broadleaf Forests                                 -4.0170519
## Urban and Built-up Lands-Evergreen Broadleaf Forests                 -6.4876162
## Water Bodies-Evergreen Broadleaf Forests                             -7.7696717
## Woody Savannas-Evergreen Broadleaf Forests                           -3.8468664
## Grasslands-Evergreen Needleleaf Forests                              -1.1148976
## Open Shrublands-Evergreen Needleleaf Forests                          3.1929062
## Permanent Snow and Ice-Evergreen Needleleaf Forests                  -7.0814981
## Permanent Wetlands-Evergreen Needleleaf Forests                      -6.1240290
## Savannas-Evergreen Needleleaf Forests                                -1.5534836
## Urban and Built-up Lands-Evergreen Needleleaf Forests                -4.0240479
## Water Bodies-Evergreen Needleleaf Forests                            -5.3061034
## Woody Savannas-Evergreen Needleleaf Forests                          -1.3832982
## Open Shrublands-Grasslands                                            4.3078037
## Permanent Snow and Ice-Grasslands                                    -5.9666005
## Permanent Wetlands-Grasslands                                        -5.0091315
## Savannas-Grasslands                                                  -0.4385861
## Urban and Built-up Lands-Grasslands                                  -2.9091503
## Water Bodies-Grasslands                                              -4.1912059
## Woody Savannas-Grasslands                                            -0.2684006
## Permanent Snow and Ice-Open Shrublands                              -10.2744043
## Permanent Wetlands-Open Shrublands                                   -9.3169352
## Savannas-Open Shrublands                                             -4.7463898
## Urban and Built-up Lands-Open Shrublands                             -7.2169541
## Water Bodies-Open Shrublands                                         -8.4990096
## Woody Savannas-Open Shrublands                                       -4.5762043
## Permanent Wetlands-Permanent Snow and Ice                             0.9574691
## Savannas-Permanent Snow and Ice                                       5.5280145
## Urban and Built-up Lands-Permanent Snow and Ice                       3.0574502
## Water Bodies-Permanent Snow and Ice                                   1.7753947
## Woody Savannas-Permanent Snow and Ice                                 5.6981999
## Savannas-Permanent Wetlands                                           4.5705454
## Urban and Built-up Lands-Permanent Wetlands                           2.0999811
## Water Bodies-Permanent Wetlands                                       0.8179256
## Woody Savannas-Permanent Wetlands                                     4.7407309
## Urban and Built-up Lands-Savannas                                    -2.4705643
## Water Bodies-Savannas                                                -3.7526198
## Woody Savannas-Savannas                                               0.1701855
## Water Bodies-Urban and Built-up Lands                                -1.2820555
## Woody Savannas-Urban and Built-up Lands                               2.6407497
## Woody Savannas-Water Bodies                                           3.9228052
```

```
##                                                                    lwr
## Closed Shrublands-Barren                                      5.08719557
## Cropland/Natural Vegetation Mosaics-Barren                  -10.01575760
## Croplands-Barren                                             -1.73887738
## Evergreen Broadleaf Forests-Barren                           -6.01568380
## Evergreen Needleleaf Forests-Barren                           1.76309301
## Grasslands-Barren                                             1.26817184
## Open Shrublands-Barren                                        3.93631577
## Permanent Snow and Ice-Barren                               -20.31722785
## Permanent Wetlands-Barren                                   -16.45916635
## Savannas-Barren                                               0.06735998
## Urban and Built-up Lands-Barren                             -10.66235422
## Water Bodies-Barren                                         -24.03361269
## Woody Savannas-Barren                                         0.23754545
## Cropland/Natural Vegetation Mosaics-Closed Shrublands       -20.58565103
## Croplands-Closed Shrublands                                 -12.45749348
## Evergreen Broadleaf Forests-Closed Shrublands               -16.58278046
## Evergreen Needleleaf Forests-Closed Shrublands               -8.92745298
## Grasslands-Closed Shrublands                                 -9.44960060
## Open Shrublands-Closed Shrublands                            -6.72211995
## Permanent Snow and Ice-Closed Shrublands                    -30.87076546
## Permanent Wetlands-Closed Shrublands                        -27.02005251
## Savannas-Closed Shrublands                                  -10.61790460
## Urban and Built-up Lands-Closed Shrublands                  -21.23736411
## Water Bodies-Closed Shrublands                              -34.57789637
## Woody Savannas-Closed Shrublands                            -10.44771914
## Croplands-Cropland/Natural Vegetation Mosaics               -15.47430531
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics -14.96440860
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics -11.53069688
## Grasslands-Cropland/Natural Vegetation Mosaics              -12.45535006
## Open Shrublands-Cropland/Natural Vegetation Mosaics          -8.70433946
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics  -28.28803038
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics      -24.97931410
## Savannas-Cropland/Natural Vegetation Mosaics                -13.13138422
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics -20.11219568
## Water Bodies-Cropland/Natural Vegetation Mosaics            -31.24611396
## Woody Savannas-Cropland/Natural Vegetation Mosaics          -12.96119875
## Evergreen Broadleaf Forests-Croplands                        -7.70769724
## Evergreen Needleleaf Forests-Croplands                        1.39462132
## Grasslands-Croplands                                          1.49904927
## Open Shrublands-Croplands                                     3.11702453
## Permanent Snow and Ice-Croplands                            -22.11871890
## Permanent Wetlands-Croplands                                -18.20162621
## Savannas-Croplands                                           -0.38676258
## Urban and Built-up Lands-Croplands                          -12.28921976
## Water Bodies-Croplands                                      -25.90860138
## Woody Savannas-Croplands                                     -0.21657712
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests    -16.95863007
## Grasslands-Evergreen Broadleaf Forests                      -17.89286183
## Open Shrublands-Evergreen Broadleaf Forests                 -14.11449220
## Permanent Snow and Ice-Evergreen Broadleaf Forests          -33.42581031
## Permanent Wetlands-Evergreen Broadleaf Forests              -30.16403819
## Savannas-Evergreen Broadleaf Forests                        -18.55697860
## Urban and Built-up Lands-Evergreen Broadleaf Forests        -25.36700194
```

```
## Water Bodies-Evergreen Broadleaf Forests                          -36.31262006
## Woody Savannas-Evergreen Broadleaf Forests                        -18.38679313
## Grasslands-Evergreen Needleleaf Forests                            -3.88235109
## Open Shrublands-Evergreen Needleleaf Forests                       -1.61294003
## Permanent Snow and Ice-Evergreen Needleleaf Forests               -26.39591752
## Permanent Wetlands-Evergreen Needleleaf Forests                   -22.50388458
## Savannas-Evergreen Needleleaf Forests                              -5.31730273
## Urban and Built-up Lands-Evergreen Needleleaf Forests             -16.64102350
## Water Bodies-Evergreen Needleleaf Forests                         -30.15478082
## Woody Savannas-Evergreen Needleleaf Forests                        -5.14711726
## Open Shrublands-Grasslands                                          0.07799593
## Permanent Snow and Ice-Grasslands                                 -25.14580579
## Permanent Wetlands-Grasslands                                     -21.22932827
## Savannas-Grasslands                                                -3.43215426
## Urban and Built-up Lands-Grasslands                               -15.31814666
## Water Bodies-Grasslands                                           -28.93493001
## Woody Savannas-Grasslands                                          -3.26196879
## Permanent Snow and Ice-Open Shrublands                            -29.85192382
## Permanent Wetlands-Open Shrublands                                -26.00621745
## Savannas-Open Shrublands                                           -9.68590485
## Urban and Built-up Lands-Open Shrublands                          -20.23311893
## Water Bodies-Open Shrublands                                      -33.55273670
## Woody Savannas-Open Shrublands                                     -9.51571938
## Permanent Wetlands-Permanent Snow and Ice                         -24.11208982
## Savannas-Permanent Snow and Ice                                   -13.82009783
## Urban and Built-up Lands-Permanent Snow and Ice                   -19.73226653
## Water Bodies-Permanent Snow and Ice                               -29.49183870
## Woody Savannas-Permanent Snow and Ice                             -13.64991237
## Savannas-Permanent Wetlands                                       -11.84902581
## Urban and Built-up Lands-Permanent Wetlands                       -18.26233842
## Water Bodies-Permanent Wetlands                                   -28.72683284
## Woody Savannas-Permanent Wetlands                                 -11.67884034
## Urban and Built-up Lands-Savannas                                 -15.13905764
## Water Bodies-Savannas                                             -28.62749505
## Woody Savannas-Savannas                                            -3.76287765
## Water Bodies-Urban and Built-up Lands                             -28.91864646
## Woody Savannas-Urban and Built-up Lands                           -10.02774366
## Woody Savannas-Water Bodies                                       -20.95207004
##                                                                          upr
## Closed Shrublands-Barren                                          15.9355064
## Cropland/Natural Vegetation Mosaics-Barren                        18.0694277
## Croplands-Barren                                                   6.0031068
## Evergreen Broadleaf Forests-Barren                                23.4842759
## Evergreen Needleleaf Forests-Barren                               10.7783626
## Grasslands-Barren                                                  9.0434886
## Open Shrublands-Barren                                            14.9909521
## Permanent Snow and Ice-Barren                                     18.6956873
## Permanent Wetlands-Barren                                         16.7525639
## Savannas-Barren                                                    9.3671283
## Urban and Built-up Lands-Barren                                   15.1557140
## Water Bodies-Barren                                               25.9628615
## Woody Savannas-Barren                                              9.5373138
## Cropland/Natural Vegetation Mosaics-Closed Shrublands              7.6166191
## Croplands-Closed Shrublands                                       -4.3009791
```

```
## Evergreen Broadleaf Forests-Closed Shrublands                      13.0286706
## Evergreen Needleleaf Forests-Closed Shrublands                      0.4462066
## Grasslands-Closed Shrublands                                       -1.2614409
## Open Shrublands-Closed Shrublands                                   4.6266858
## Permanent Snow and Ice-Closed Shrublands                            8.2265229
## Permanent Wetlands-Closed Shrublands                                6.2907480
## Savannas-Closed Shrublands                                         -0.9703091
## Urban and Built-up Lands-Closed Shrublands                          4.7080219
## Water Bodies-Closed Shrublands                                     15.4844431
## Woody Savannas-Closed Shrublands                                   -0.8001236
## Croplands-Cropland/Natural Vegetation Mosaics                      11.6848646
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics    24.3793306
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics   16.0184824
## Grasslands-Cropland/Natural Vegetation Mosaics                     14.7133404
## Open Shrublands-Cropland/Natural Vegetation Mosaics                19.5779373
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics         18.6128197
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics             17.2190415
## Savannas-Cropland/Natural Vegetation Mosaics                       14.5122024
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics       16.5518854
## Water Bodies-Cropland/Natural Vegetation Mosaics                   25.1216926
## Woody Savannas-Cropland/Natural Vegetation Mosaics                 14.6823879
## Evergreen Broadleaf Forests-Croplands                             20.9120599
## Evergreen Needleleaf Forests-Croplands                             6.8826049
## Grasslands-Croplands                                               4.5483818
## Open Shrublands-Croplands                                         11.5460140
## Permanent Snow and Ice-Croplands                                  16.2329489
## Permanent Wetlands-Croplands                                      14.2307943
## Savannas-Croplands                                                 5.5570215
## Urban and Built-up Lands-Croplands                                12.5183502
## Water Bodies-Croplands                                            23.5736208
## Woody Savannas-Croplands                                           5.7272070
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests          12.0314936
## Grasslands-Evergreen Broadleaf Forests                            10.7359302
## Open Shrublands-Evergreen Broadleaf Forests                       15.5731680
## Permanent Snow and Ice-Evergreen Broadleaf Forests                14.3356776
## Permanent Wetlands-Evergreen Broadleaf Forests                    12.9888436
## Savannas-Evergreen Broadleaf Forests                              10.5228748
## Urban and Built-up Lands-Evergreen Broadleaf Forests              12.3917696
## Water Bodies-Evergreen Broadleaf Forests                          20.7732767
## Woody Savannas-Evergreen Broadleaf Forests                        10.6930603
## Grasslands-Evergreen Needleleaf Forests                            1.6525560
## Open Shrublands-Evergreen Needleleaf Forests                       7.9987523
## Permanent Snow and Ice-Evergreen Needleleaf Forests               12.2329213
## Permanent Wetlands-Evergreen Needleleaf Forests                   10.2558265
## Savannas-Evergreen Needleleaf Forests                              2.2103354
## Urban and Built-up Lands-Evergreen Needleleaf Forests              8.5929277
## Water Bodies-Evergreen Needleleaf Forests                         19.5425740
## Woody Savannas-Evergreen Needleleaf Forests                        2.3805209
## Open Shrublands-Grasslands                                         8.5376115
## Permanent Snow and Ice-Grasslands                                 13.2126047
## Permanent Wetlands-Grasslands                                     11.2110653
## Savannas-Grasslands                                                2.5549821
## Urban and Built-up Lands-Grasslands                                9.4998460
## Water Bodies-Grasslands                                           20.5525183
```

```
## Woody Savannas-Grasslands                                          2.7251676
## Permanent Snow and Ice-Open Shrublands                             9.3031153
## Permanent Wetlands-Open Shrublands                                 7.3723471
## Savannas-Open Shrublands                                           0.1931253
## Urban and Built-up Lands-Open Shrublands                           5.7992108
## Water Bodies-Open Shrublands                                      16.5547176
## Woody Savannas-Open Shrublands                                     0.3633107
## Permanent Wetlands-Permanent Snow and Ice                         26.0270279
## Savannas-Permanent Snow and Ice                                   24.8761267
## Urban and Built-up Lands-Permanent Snow and Ice                   25.8471669
## Water Bodies-Permanent Snow and Ice                               33.0426281
## Woody Savannas-Permanent Snow and Ice                             25.0463122
## Savannas-Permanent Wetlands                                       20.9901166
## Urban and Built-up Lands-Permanent Wetlands                       22.4623007
## Water Bodies-Permanent Wetlands                                   30.3626841
## Woody Savannas-Permanent Wetlands                                 21.1603021
## Urban and Built-up Lands-Savannas                                 10.1979291
## Water Bodies-Savannas                                             21.1222555
## Woody Savannas-Savannas                                            4.1032486
## Water Bodies-Urban and Built-up Lands                             26.3545354
## Woody Savannas-Urban and Built-up Lands                           15.3092431
## Woody Savannas-Water Bodies                                       28.7976805
##                                                                       p adj
## Closed Shrublands-Barren                                          0.0000000
## Cropland/Natural Vegetation Mosaics-Barren                        0.9995602
## Croplands-Barren                                                  0.8538217
## Evergreen Broadleaf Forests-Barren                                0.7761403
## Evergreen Needleleaf Forests-Barren                               0.0002695
## Grasslands-Barren                                                 0.0007390
## Open Shrublands-Barren                                            0.0000009
## Permanent Snow and Ice-Barren                                     1.0000000
## Permanent Wetlands-Barren                                         1.0000000
## Savannas-Barren                                                   0.0428685
## Urban and Built-up Lands-Barren                                   0.9999987
## Water Bodies-Barren                                               1.0000000
## Woody Savannas-Barren                                             0.0286289
## Cropland/Natural Vegetation Mosaics-Closed Shrublands             0.9596610
## Croplands-Closed Shrublands                                       0.0000000
## Evergreen Broadleaf Forests-Closed Shrublands                     1.0000000
## Evergreen Needleleaf Forests-Closed Shrublands                    0.1259348
## Grasslands-Closed Shrublands                                      0.0009697
## Open Shrublands-Closed Shrublands                                 0.9999973
## Permanent Snow and Ice-Closed Shrublands                          0.8023590
## Permanent Wetlands-Closed Shrublands                              0.7097379
## Savannas-Closed Shrublands                                        0.0044328
## Urban and Built-up Lands-Closed Shrublands                        0.6748861
## Water Bodies-Closed Shrublands                                    0.9920502
## Woody Savannas-Closed Shrublands                                  0.0070755
## Croplands-Cropland/Natural Vegetation Mosaics                     0.9999999
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics   0.9999417
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics  0.9999994
## Grasslands-Cropland/Natural Vegetation Mosaics                    1.0000000
## Open Shrublands-Cropland/Natural Vegetation Mosaics               0.9914325
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics        0.9999897
```

```
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics           0.9999974
## Savannas-Cropland/Natural Vegetation Mosaics                     1.0000000
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics     1.0000000
## Water Bodies-Cropland/Natural Vegetation Mosaics                 1.0000000
## Woody Savannas-Cropland/Natural Vegetation Mosaics               1.0000000
## Evergreen Broadleaf Forests-Croplands                           0.9585941
## Evergreen Needleleaf Forests-Croplands                          0.0000384
## Grasslands-Croplands                                            0.0000000
## Open Shrublands-Croplands                                       0.0000006
## Permanent Snow and Ice-Croplands                                0.9999997
## Permanent Wetlands-Croplands                                    1.0000000
## Savannas-Croplands                                              0.1698038
## Urban and Built-up Lands-Croplands                              1.0000000
## Water Bodies-Croplands                                          1.0000000
## Woody Savannas-Croplands                                        0.1028402
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests        0.9999990
## Grasslands-Evergreen Broadleaf Forests                          0.9999040
## Open Shrublands-Evergreen Broadleaf Forests                     1.0000000
## Permanent Snow and Ice-Evergreen Broadleaf Forests              0.9877679
## Permanent Wetlands-Evergreen Broadleaf Forests                  0.9882277
## Savannas-Evergreen Broadleaf Forests                            0.9997070
## Urban and Built-up Lands-Evergreen Broadleaf Forests            0.9970990
## Water Bodies-Evergreen Broadleaf Forests                        0.9997512
## Woody Savannas-Evergreen Broadleaf Forests                      0.9998185
## Grasslands-Evergreen Needleleaf Forests                         0.9868680
## Open Shrublands-Evergreen Needleleaf Forests                    0.6078893
## Permanent Snow and Ice-Evergreen Needleleaf Forests             0.9945290
## Permanent Wetlands-Evergreen Needleleaf Forests                 0.9934086
## Savannas-Evergreen Needleleaf Forests                           0.9837108
## Urban and Built-up Lands-Evergreen Needleleaf Forests           0.9986413
## Water Bodies-Evergreen Needleleaf Forests                       0.9999845
## Woody Savannas-Evergreen Needleleaf Forests                     0.9944014
## Open Shrublands-Grasslands                                      0.0410813
## Permanent Snow and Ice-Grasslands                               0.9989518
## Permanent Wetlands-Grasslands                                   0.9990296
## Savannas-Grasslands                                             0.9999998
## Urban and Built-up Lands-Grasslands                             0.9999539
## Water Bodies-Grasslands                                         0.9999990
## Woody Savannas-Grasslands                                       1.0000000
## Permanent Snow and Ice-Open Shrublands                          0.8932648
## Permanent Wetlands-Open Shrublands                              0.8411087
## Savannas-Open Shrublands                                        0.0744495
## Urban and Built-up Lands-Open Shrublands                        0.8476512
## Water Bodies-Open Shrublands                                    0.9974503
## Woody Savannas-Open Shrublands                                  0.1034871
## Permanent Wetlands-Permanent Snow and Ice                       1.0000000
## Savannas-Permanent Snow and Ice                                 0.9995773
## Urban and Built-up Lands-Permanent Snow and Ice                 0.9999999
## Water Bodies-Permanent Snow and Ice                             1.0000000
## Woody Savannas-Permanent Snow and Ice                           0.9994137
## Savannas-Permanent Wetlands                                     0.9996819
## Urban and Built-up Lands-Permanent Wetlands                     1.0000000
## Water Bodies-Permanent Wetlands                                 1.0000000
## Woody Savannas-Permanent Wetlands                               0.9995265
```

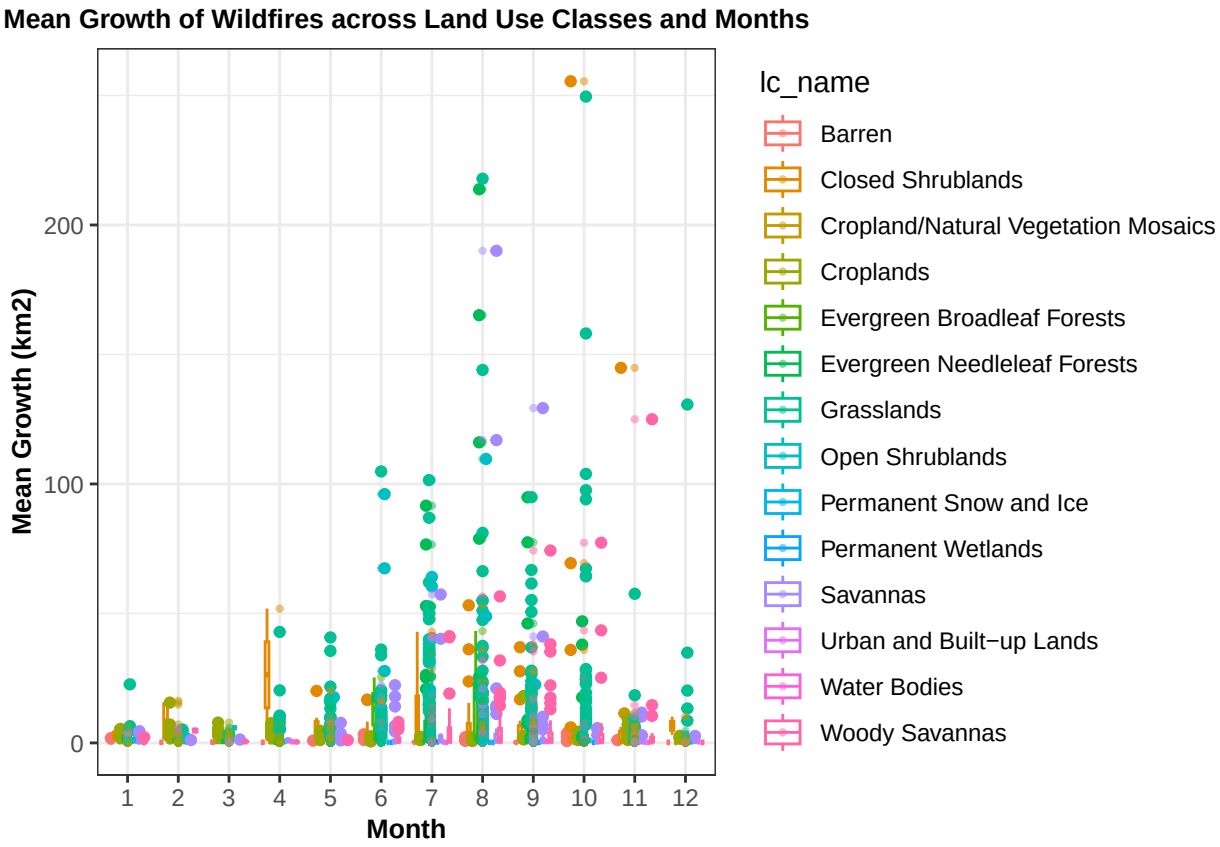
```
## Urban and Built-up Lands-Savannas                                0.9999947
## Water Bodies-Savannas                                            0.9999998
## Woody Savannas-Savannas                                          1.0000000
## Water Bodies-Urban and Built-up Lands                            1.0000000
## Woody Savannas-Urban and Built-up Lands                          0.9999883
## Woody Savannas-Water Bodies                                      0.9999996
```

```
#Sorting the Tukey tests so they are more readable
mean_growth_aov2_sort <- as.data.frame(TukeyHSD(mean_growth_aov2)$lc_name) |>
  filter(`p adj` < 0.05) |>
  arrange(`p adj`)
```

Figures

```
#Plot of mean growth for each land cover class and month to address the question,
#"What is the relationship between time of year and which land cover types see the
#fastest fires?" The x-axis shows the month of the year and the y-axis shows the
#mean growth per day in km2.
mean_growth_anova_plot <- ggplot(mean_growth, aes(y = mu_grw_km2, x = ig_month,
                                                  color = lc_name)) +
  geom_boxplot() +
  geom_point(alpha = 0.5, size= 0.8) +
  labs(title = "Mean Growth of Wildfires across Land Use Classes and Months",
       x= "Month",
       y="Mean Growth (km2)")

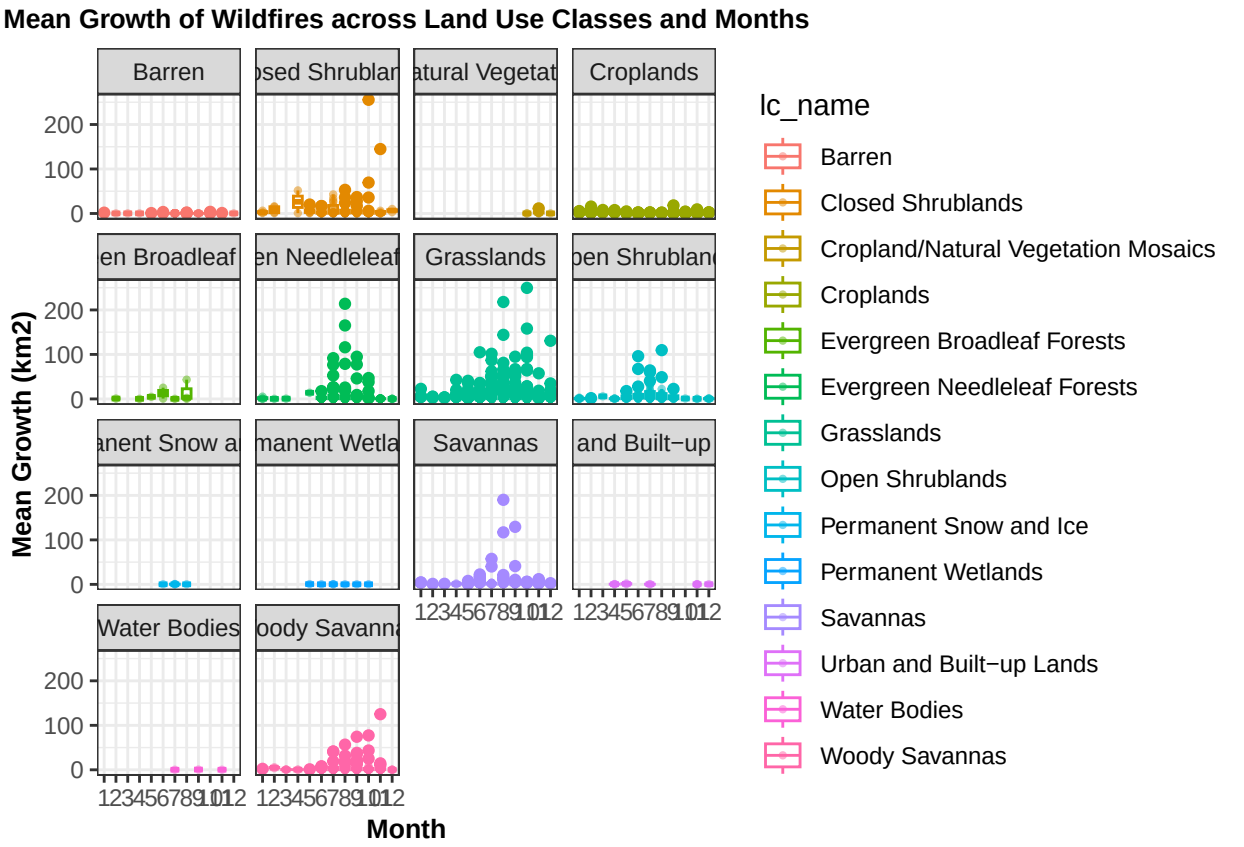
mean_growth_anova_plot
```




```
#Key finding: The mean growth per day seems to be in the late summer to fall (August - October).

#Plot of mean growth for each month separated by land cover class to address the question,
#"What is the relationship between time of year and which land cover types see the fastest
#fires?" The x-axis shows the month of the year and the y-axis shows the mean growth per
#day in km2.
mean_growth_anova_plot_separate <- ggplot(mean_growth, aes(y = mu_grw_km2, x = ig_month,
                                                           color = lc_name)) +
  geom_boxplot() +
  geom_point(alpha = 0.5, size= 0.8) +
  facet_wrap(vars(lc_name)) +
  labs(title = "Mean Growth of Wildfires across Land Use Classes and Months",
       x= "Month",
       y="Mean Growth (km2)")

mean_growth_anova_plot_separate
```




```
#Key finding: Land cover does make a difference in the fire growth per day. Water bodies,
#permanent wetlands, and permanent snow and ice see very modest mean fire growth per day
#while closed shrublands, evergreen needleleaf forests, grasslands, open shrublands,
#savannas, and woody savannas all have higher mean growth periods and greater variability
#based on time of year.
```

###Analysis

To begin to understand if there is a relationship between time of year and which land cover types experience the fastest fires, we completed a two way ANOVA as well as Tukey's Honest Significant Differences. The null hypothesis is that there is no difference between time of year or land cover type with mean fire growth. P-values <.05 mean that the null hypothesis can be rejected.

Tukey's Honest Significant Differences showed significant difference in mean growth rate between the following land cover classes: Crops-Closed Shrubland (Closed Shrublands is higher, 0.00000004094677), Grasslands-Croplands (Grasslands is higher, 0.00000004328649), Closed Shrublands-Barren (Closed Shrublands is higher, 0.00000004836993), Open Shrublands-Croplands (Open Shrublands is higher, 0.0000005551389), Open Shrublands-Barren (Open Shrublands is higher, 0.0000009260920), Evergreen Needleleaf Forests-Croplands (Evergreen Needleleaf Forests is higher, 0.00003844067), Evergreen Needleleaf Forests-Barren (Evergreen Needleleaf Forests is higher, 0.000269493), Grasslands-Barren (Grasslands is higher, 0.0007389839), Grasslands-Closed Shrublands (Closed Shrublands is higher, 0.0009696893), Savannas-Closed Shrublands (Closed Shrublands is higher, 0.004432793), Woody Savannas-Closed Shrublands (Closed Shrublands is higher, 0.007075468), Woody Savannas-Barren (Woody Savannas is higher, 0.02862893), Open Shrublands-Grasslands (Open Shrublands is higher, 0.04108133), and Savannas-Barren (Savannas is higher, 0.04286845). Closed shrublands and open shrublands seem to be particularly significant in meant growth of fires based on these results.

Additionally, Tukey's Honest Significant Differences showed significant difference in mean growth rate between the following months: August and January (August is higher, 0.0001989), July and February (July is higher, 0.0016073), August and February (August is higher, 0.0000019), July and March (July is higher, 0.0047414), August and March (August is higher, 0.0000127), July and April (July is higher, 0.0143088), August and April (August is higher, 0.0000339), July and May (July is higher, 0.0001859), August and May (August is higher, 0.0000001), July and June (July is higher, 0.0133577), August and June (August is higher, 0.0000031), November and July (July is higher, 0.0000337), September and August (August is higher, 0.0001458), October and August (August is higher, 0.0286293), November and August (August is higher, 0.0000000), December and August (August is higher, 0.0395852), and November and October (October is higher, 0.0341653). These results indicate that July and August are particulalry meaningful in the mean growth of fires.

Something to consider is this result from the linear mode: multiple R-squared is 0.05292 and adjusted R-squared is 0.04764, so these variables explain very little of the variation.

What is the relationship between time of year and which land cover type sees the longest fast fires?

Response variable: event length. Predictors: land cover, month, year.

```
# Group data
event_length_group <- california_events_processed %>%
  group_by(lc_name, ig_month, ig_year) %>%
  summarise(mean_event_dur = mean(event_dur),
            max_event_dur = max(event_dur),
            min_event_dur = min(event_dur)) %>%
  ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by lc_name, ig_month, and ig_year.
## i Output is grouped by lc_name and ig_month.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(lc_name, ig_month, ig_year))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```
# create new data frame of just question variables
event_length <- california_events_processed %>%
  select(lc_name, ig_month, ig_year, event_dur)
```

Two-way ANOVA

```
# Format as aov
eventlength_aov <- aov(data = event_length, event_dur ~ ig_month + lc_name)
summary(eventlength_aov) # The null hypothesis is that there is no difference in time of year or
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## ig_month      11  16667  1515.2   20.11 <2e-16 ***
## lc_name       13  21518  1655.2   21.97 <2e-16 ***
## Residuals   4304 324270    75.3
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#land cover type in their relationship to fast fire event length. P-values <.05 mean that the null
#hypothesis can be rejected and that there is a difference between at least two months and at
#least two land cover types.

# Run a post-hoc test to look for pairwise differences
TukeyHSD(eventlength_aov)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = event_dur ~ ig_month + lc_name, data = event_length)
##
## $ig_month
##               diff         lwr        upr     p adj
## 2-1   -0.485794378 -3.66390897  2.6923202 0.9999976
## 3-1    2.557274975 -0.75230661  5.8668566 0.3233822
## 4-1   -1.749503713 -4.90854717  1.4095397 0.8123642
## 5-1   -1.999813898 -4.83181471  0.8321869 0.4701749
## 6-1   -0.479251143 -3.22943406  2.2709318 0.9999906
## 7-1    1.341722944 -1.40470219  4.0881481 0.9103993
## 8-1    0.722919147 -2.04452117  3.4903595 0.9994717
## 9-1    4.182222449  1.46362774  6.9008172 0.0000328
## 10-1   0.071907788 -2.68608680  2.8299024 1.0000000
## 11-1  -1.985737977 -4.74956839  0.7780924 0.4414364
## 12-1  -1.437458724 -5.06237453  2.1874571 0.9796933
## 3-2    3.043069353  0.03792455  6.0482142 0.0439883
## 4-2   -1.263709335 -4.10221609  1.5747974 0.9520446
## 5-2   -1.514019520 -3.98339062  0.9553516 0.6903190
## 6-2    0.006543235 -2.36855068  2.3816372 1.0000000
## 7-2    1.827517323 -0.54322434  4.1982590 0.3270778
## 8-2    1.208713525 -1.18634198  3.6037690 0.8901621
## 9-2    4.668016828  2.32957243  7.0064612 0.0000000
## 10-2   0.557702166 -1.82643272  2.9418370 0.9998180
## 11-2  -1.499943598 -3.89082703  0.8909398 0.6577322
## 12-2  -0.951664346 -4.30092988  2.3976012 0.9988371
## 4-3   -4.306778688 -7.29174744 -1.3218099 0.0001554
## 5-3   -4.557088873 -7.19350908 -1.9206687 0.0000011
## 6-3   -3.036526118 -5.58485675 -0.4881955 0.0056019
## 7-3   -1.215552031 -3.75982676  1.3287227 0.9224471
## 8-3   -1.834355828 -4.40130125  0.7325896 0.4503971
## 9-3    1.624947475 -0.88926017  4.1391551 0.6131938
## 10-3  -2.485367187 -5.04212627  0.0713919 0.0658162
## 11-3  -4.543012952 -7.10606611 -1.9799598 0.0000005
## 12-3  -3.994733699 -7.46899585 -0.5204715 0.0094241
## 5-4   -0.250310185 -2.69508764  2.1944673 1.0000000
## 6-4    1.270252570 -1.07926106  3.6197662 0.8355279
## 7-4    3.091226657  0.74611275  5.4363406 0.0010115
## 8-4    2.472422860  0.10273214  4.8421136 0.0318828
## 9-4    5.931726162  3.61926743  8.2441849 0.0000000
## 10-4   1.821411501 -0.53724115  4.1800642 0.3243174
## 11-4  -0.236234264 -2.60170817  2.1292396 1.0000000
## 12-4   0.312044989 -3.01912940  3.6432194 1.0000000
## 6-5    1.520562755 -0.36646860  3.4075941 0.2603841
```

```
## 7-5    3.341536842  1.45998635  5.2230873 0.0000005
## 8-5    2.722733045  0.81063809  4.6348280 0.0002087
## 9-5    6.182036347  4.34134668  8.0227260 0.0000000
## 10-5   2.071721686  0.17332358  3.9701198 0.0187772
## 11-5   0.014075921 -1.89279057  1.9209424 1.0000000
## 12-5   0.562355174 -2.46045740  3.5851677 0.9999817
## 7-6    1.820974087  0.06498156  3.5769666 0.0341129
## 8-6    1.202170290 -0.58651208  2.9908527 0.5514097
## 9-6    4.661473593  2.94933577  6.3736114 0.0000000
## 10-6   0.551158931 -1.22287401  2.3251919 0.9973801
## 11-6  -1.506486834 -3.28957890  0.2766052 0.1965911
## 12-6  -0.958207581 -3.90450582  1.9880907 0.9960740
## 8-7   -0.618803798 -2.40170299  1.1640954 0.9931281
## 9-7    2.840499505  1.13440430  4.5465947 0.0000036
## 10-7  -1.269815156 -3.03801700  0.4983867 0.4422080
## 11-7  -3.327460921 -5.10475162 -1.5501702 0.0000001
## 12-7  -2.779181668 -5.72197256  0.1636092 0.0853990
## 9-8    3.459303303  1.71958042  5.1990262 0.0000000
## 10-8  -0.651011359 -2.45168141  1.1496587 0.9903306
## 11-8  -2.708657123 -4.51825296 -0.8990613 0.0000653
## 12-8  -2.160377871 -5.12279126  0.8020355 0.4167442
## 10-9  -4.110314662 -5.83497230 -2.3856570 0.0000000
## 11-9  -6.167960426 -7.90193517 -4.4339857 0.0000000
## 12-9  -5.619681173 -8.53651583 -2.7028465 0.0000001
## 11-10 -2.057645765 -3.85276285 -0.2625287 0.0098410
## 12-10 -1.509366512 -4.46295778  1.4442248 0.8814594
## 12-11  0.548279253 -2.41076211  3.5073206 0.9999824
##
## $lc_name
##                                                                    diff
## Closed Shrublands-Barren                                      0.62173414
## Cropland/Natural Vegetation Mosaics-Barren                    3.52890766
## Croplands-Barren                                              2.42830930
## Evergreen Broadleaf Forests-Barren                            8.69654164
## Evergreen Needleleaf Forests-Barren                           8.27626474
## Grasslands-Barren                                            -0.16593464
## Open Shrublands-Barren                                        0.02142403
## Permanent Snow and Ice-Barren                                -4.34424061
## Permanent Wetlands-Barren                                    -3.51639184
## Savannas-Barren                                               0.87389296
## Urban and Built-up Lands-Barren                               0.25495207
## Water Bodies-Barren                                          -5.00320905
## Woody Savannas-Barren                                         1.49193969
## Cropland/Natural Vegetation Mosaics-Closed Shrublands         2.90717352
## Croplands-Closed Shrublands                                   1.80657516
## Evergreen Broadleaf Forests-Closed Shrublands                 8.07480750
## Evergreen Needleleaf Forests-Closed Shrublands                7.65453060
## Grasslands-Closed Shrublands                                 -0.78766878
## Open Shrublands-Closed Shrublands                            -0.60031010
## Permanent Snow and Ice-Closed Shrublands                     -4.96597475
## Permanent Wetlands-Closed Shrublands                         -4.13812598
## Savannas-Closed Shrublands                                    0.25215883
## Urban and Built-up Lands-Closed Shrublands                   -0.36678207
## Water Bodies-Closed Shrublands                               -5.62494318
```

```
## Woody Savannas-Closed Shrublands                                          0.87020555
## Croplands-Cropland/Natural Vegetation Mosaics                            -1.10059836
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics           5.16763398
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics          4.74735708
## Grasslands-Cropland/Natural Vegetation Mosaics                           -3.69484230
## Open Shrublands-Cropland/Natural Vegetation Mosaics                      -3.50748362
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics               -7.87314827
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics                   -7.04529950
## Savannas-Cropland/Natural Vegetation Mosaics                             -2.65501469
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics             -3.27395559
## Water Bodies-Cropland/Natural Vegetation Mosaics                         -8.53211670
## Woody Savannas-Cropland/Natural Vegetation Mosaics                       -2.03696797
## Evergreen Broadleaf Forests-Croplands                                    6.26823234
## Evergreen Needleleaf Forests-Croplands                                   5.84795544
## Grasslands-Croplands                                                    -2.59424393
## Open Shrublands-Croplands                                               -2.40688526
## Permanent Snow and Ice-Croplands                                        -6.77254991
## Permanent Wetlands-Croplands                                            -5.94470114
## Savannas-Croplands                                                      -1.55441633
## Urban and Built-up Lands-Croplands                                      -2.17335723
## Water Bodies-Croplands                                                  -7.43151834
## Woody Savannas-Croplands                                                -0.93636961
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests                -0.42027690
## Grasslands-Evergreen Broadleaf Forests                                  -8.86247627
## Open Shrublands-Evergreen Broadleaf Forests                             -8.67511760
## Permanent Snow and Ice-Evergreen Broadleaf Forests                     -13.04078224
## Permanent Wetlands-Evergreen Broadleaf Forests                         -12.21293348
## Savannas-Evergreen Broadleaf Forests                                    -7.82264867
## Urban and Built-up Lands-Evergreen Broadleaf Forests                    -8.44158957
## Water Bodies-Evergreen Broadleaf Forests                               -13.69975068
## Woody Savannas-Evergreen Broadleaf Forests                              -7.20460195
## Grasslands-Evergreen Needleleaf Forests                                 -8.44219938
## Open Shrublands-Evergreen Needleleaf Forests                            -8.25484071
## Permanent Snow and Ice-Evergreen Needleleaf Forests                    -12.62050535
## Permanent Wetlands-Evergreen Needleleaf Forests                        -11.79265658
## Savannas-Evergreen Needleleaf Forests                                   -7.40237178
## Urban and Built-up Lands-Evergreen Needleleaf Forests                   -8.02131267
## Water Bodies-Evergreen Needleleaf Forests                              -13.27947379
## Woody Savannas-Evergreen Needleleaf Forests                             -6.78432505
## Open Shrublands-Grasslands                                               0.18735867
## Permanent Snow and Ice-Grasslands                                       -4.17830597
## Permanent Wetlands-Grasslands                                           -3.35045721
## Savannas-Grasslands                                                      1.03982760
## Urban and Built-up Lands-Grasslands                                      0.42088670
## Water Bodies-Grasslands                                                 -4.83727441
## Woody Savannas-Grasslands                                                1.65787433
## Permanent Snow and Ice-Open Shrublands                                  -4.36566464
## Permanent Wetlands-Open Shrublands                                      -3.53781588
## Savannas-Open Shrublands                                                 0.85246893
## Urban and Built-up Lands-Open Shrublands                                 0.23352803
## Water Bodies-Open Shrublands                                            -5.02463308
## Woody Savannas-Open Shrublands                                           1.47051566
## Permanent Wetlands-Permanent Snow and Ice                                0.82784876
## Savannas-Permanent Snow and Ice                                          5.21813357
```

```
## Urban and Built-up Lands-Permanent Snow and Ice                    4.59919267
## Water Bodies-Permanent Snow and Ice                               -0.65896844
## Woody Savannas-Permanent Snow and Ice                              5.83618030
## Savannas-Permanent Wetlands                                        4.39028481
## Urban and Built-up Lands-Permanent Wetlands                        3.77134391
## Water Bodies-Permanent Wetlands                                   -1.48681720
## Woody Savannas-Permanent Wetlands                                  5.00833154
## Urban and Built-up Lands-Savannas                                 -0.61894090
## Water Bodies-Savannas                                             -5.87710201
## Woody Savannas-Savannas                                            0.61804673
## Water Bodies-Urban and Built-up Lands                             -5.25816111
## Woody Savannas-Urban and Built-up Lands                            1.23698762
## Woody Savannas-Water Bodies                                        6.49514874
##                                                                          lwr
## Closed Shrublands-Barren                                          -3.0682474
## Cropland/Natural Vegetation Mosaics-Barren                        -6.0240832
## Croplands-Barren                                                  -0.2050756
## Evergreen Broadleaf Forests-Barren                                -1.3376754
## Evergreen Needleleaf Forests-Barren                                5.2097802
## Grasslands-Barren                                                 -2.8106574
## Open Shrublands-Barren                                            -3.7387377
## Permanent Snow and Ice-Barren                                    -17.6142268
## Permanent Wetlands-Barren                                        -14.8131432
## Savannas-Barren                                                   -2.2893620
## Urban and Built-up Lands-Barren                                   -8.5268938
## Water Bodies-Barren                                              -22.0091805
## Woody Savannas-Barren                                             -1.6713153
## Cropland/Natural Vegetation Mosaics-Closed Shrublands             -6.6856430
## Croplands-Closed Shrublands                                       -0.9678095
## Evergreen Broadleaf Forests-Closed Shrublands                     -1.9973326
## Evergreen Needleleaf Forests-Closed Shrublands                     4.4661420
## Grasslands-Closed Shrublands                                      -3.5728174
## Open Shrublands-Closed Shrublands                                 -4.4605317
## Permanent Snow and Ice-Closed Shrublands                         -18.2646599
## Permanent Wetlands-Closed Shrublands                             -15.4685754
## Savannas-Closed Shrublands                                        -3.0294073
## Urban and Built-up Lands-Closed Shrublands                        -9.1919343
## Water Bodies-Closed Shrublands                                   -22.6533183
## Woody Savannas-Closed Shrublands                                  -2.4113605
## Croplands-Cropland/Natural Vegetation Mosaics                    -10.3386112
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics   -8.2148798
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics  -4.6233148
## Grasslands-Cropland/Natural Vegetation Mosaics                   -12.9360935
## Open Shrublands-Cropland/Natural Vegetation Mosaics              -13.1275138
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics       -23.8261636
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics           -21.3987923
## Savannas-Cropland/Natural Vegetation Mosaics                     -12.0577987
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics     -15.7450013
## Water Bodies-Cropland/Natural Vegetation Mosaics                 -27.7052549
## Woody Savannas-Cropland/Natural Vegetation Mosaics               -11.4397519
## Evergreen Broadleaf Forests-Croplands                             -3.4665896
## Evergreen Needleleaf Forests-Croplands                             3.9812540
## Grasslands-Croplands                                              -3.6314543
## Open Shrublands-Croplands                                         -5.2739505
```

```
## Permanent Snow and Ice-Croplands                                  -19.8176172
## Permanent Wetlands-Croplands                                      -16.9763754
## Savannas-Croplands                                                 -3.5761554
## Urban and Built-up Lands-Croplands                                -10.6114888
## Water Bodies-Croplands                                            -24.2625704
## Woody Savannas-Croplands                                           -2.9581086
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests          -10.2810765
## Grasslands-Evergreen Broadleaf Forests                            -18.6003714
## Open Shrublands-Evergreen Broadleaf Forests                       -18.7731797
## Permanent Snow and Ice-Evergreen Broadleaf Forests                -29.2865378
## Permanent Wetlands-Evergreen Broadleaf Forests                    -26.8911021
## Savannas-Evergreen Broadleaf Forests                              -17.7139693
## Urban and Built-up Lands-Evergreen Broadleaf Forests              -21.2849871
## Water Bodies-Evergreen Broadleaf Forests                          -33.1171426
## Woody Savannas-Evergreen Broadleaf Forests                        -17.0959226
## Grasslands-Evergreen Needleleaf Forests                           -10.3248616
## Open Shrublands-Evergreen Needleleaf Forests                      -11.5241946
## Permanent Snow and Ice-Evergreen Needleleaf Forests               -25.7598505
## Permanent Wetlands-Evergreen Needleleaf Forests                   -22.9356566
## Savannas-Evergreen Needleleaf Forests                              -9.9628483
## Urban and Built-up Lands-Evergreen Needleleaf Forests             -16.6044750
## Water Bodies-Evergreen Needleleaf Forests                         -30.1837018
## Woody Savannas-Evergreen Needleleaf Forests                        -9.3448016
## Open Shrublands-Grasslands                                         -2.6901239
## Permanent Snow and Ice-Grasslands                                 -17.2256667
## Permanent Wetlands-Grasslands                                     -14.3848435
## Savannas-Grasslands                                                -0.9966574
## Urban and Built-up Lands-Grasslands                                -8.0207901
## Water Bodies-Grasslands                                           -21.6701041
## Woody Savannas-Grasslands                                          -0.3786107
## Permanent Snow and Ice-Open Shrublands                            -17.6839934
## Permanent Wetlands-Open Shrublands                                -14.8913148
## Savannas-Open Shrublands                                           -2.5078181
## Urban and Built-up Lands-Open Shrublands                           -8.6211975
## Water Bodies-Open Shrublands                                      -22.0683537
## Woody Savannas-Open Shrublands                                     -1.8897714
## Permanent Wetlands-Permanent Snow and Ice                         -16.2266420
## Savannas-Permanent Snow and Ice                                    -7.9441324
## Urban and Built-up Lands-Permanent Snow and Ice                   -10.9043515
## Water Bodies-Permanent Snow and Ice                               -21.9296555
## Woody Savannas-Permanent Snow and Ice                              -7.3260857
## Savannas-Permanent Wetlands                                        -6.7797332
## Urban and Built-up Lands-Permanent Wetlands                       -10.0808739
## Water Bodies-Permanent Wetlands                                   -21.5857273
## Woody Savannas-Permanent Wetlands                                  -6.1616865
## Urban and Built-up Lands-Savannas                                  -9.2371501
## Water Bodies-Savannas                                             -22.7991521
## Woody Savannas-Savannas                                            -2.0575643
## Water Bodies-Urban and Built-up Lands                             -24.0589700
## Woody Savannas-Urban and Built-up Lands                            -7.3812216
## Woody Savannas-Water Bodies                                       -10.4269013
##                                                                          upr
## Closed Shrublands-Barren                                           4.3117156
## Cropland/Natural Vegetation Mosaics-Barren                        13.0818985
```

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## Croplands-Barren                                                    5.0616942
## Evergreen Broadleaf Forests-Barren                                 18.7307587
## Evergreen Needleleaf Forests-Barren                                11.3427493
## Grasslands-Barren                                                   2.4787882
## Open Shrublands-Barren                                              3.7815858
## Permanent Snow and Ice-Barren                                       8.9257456
## Permanent Wetlands-Barren                                           7.7803595
## Savannas-Barren                                                     4.0371479
## Urban and Built-up Lands-Barren                                     9.0367980
## Water Bodies-Barren                                                12.0027624
## Woody Savannas-Barren                                               4.6551947
## Cropland/Natural Vegetation Mosaics-Closed Shrublands              12.4999900
## Croplands-Closed Shrublands                                         4.5809598
## Evergreen Broadleaf Forests-Closed Shrublands                      18.1469476
## Evergreen Needleleaf Forests-Closed Shrublands                     10.8429192
## Grasslands-Closed Shrublands                                        1.9974798
## Open Shrublands-Closed Shrublands                                   3.2599115
## Permanent Snow and Ice-Closed Shrublands                            8.3327104
## Permanent Wetlands-Closed Shrublands                                7.1923235
## Savannas-Closed Shrublands                                          3.5337249
## Urban and Built-up Lands-Closed Shrublands                          8.4583701
## Water Bodies-Closed Shrublands                                     11.4034319
## Woody Savannas-Closed Shrublands                                    4.1517716
## Croplands-Cropland/Natural Vegetation Mosaics                       8.1374144
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics    18.5501478
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics   14.1180290
## Grasslands-Cropland/Natural Vegetation Mosaics                      5.5464089
## Open Shrublands-Cropland/Natural Vegetation Mosaics                 6.1125466
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics          8.0798670
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics              7.3081933
## Savannas-Cropland/Natural Vegetation Mosaics                        6.7477693
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics        9.1970901
## Water Bodies-Cropland/Natural Vegetation Mosaics                   10.6410215
## Woody Savannas-Cropland/Natural Vegetation Mosaics                  7.3658160
## Evergreen Broadleaf Forests-Croplands                              16.0030543
## Evergreen Needleleaf Forests-Croplands                              7.7146569
## Grasslands-Croplands                                               -1.5570335
## Open Shrublands-Croplands                                           0.4601800
## Permanent Snow and Ice-Croplands                                    6.2725174
## Permanent Wetlands-Croplands                                        5.0869731
## Savannas-Croplands                                                  0.4673227
## Urban and Built-up Lands-Croplands                                  6.2647743
## Water Bodies-Croplands                                              9.3995337
## Woody Savannas-Croplands                                            1.0853694
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests            9.4405228
## Grasslands-Evergreen Broadleaf Forests                              0.8754188
## Open Shrublands-Evergreen Broadleaf Forests                         1.4229445
## Permanent Snow and Ice-Evergreen Broadleaf Forests                  3.2049734
## Permanent Wetlands-Evergreen Broadleaf Forests                      2.4652351
## Savannas-Evergreen Broadleaf Forests                                2.0686720
## Urban and Built-up Lands-Evergreen Broadleaf Forests                4.4018079
## Water Bodies-Evergreen Broadleaf Forests                            5.7176412
## Woody Savannas-Evergreen Broadleaf Forests                          2.6867187
## Grasslands-Evergreen Needleleaf Forests                            -6.5595372
```

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## Open Shrublands-Evergreen Needleleaf Forests                    -4.9854868
## Permanent Snow and Ice-Evergreen Needleleaf Forests              0.5188398
## Permanent Wetlands-Evergreen Needleleaf Forests                 -0.6496566
## Savannas-Evergreen Needleleaf Forests                           -4.8418952
## Urban and Built-up Lands-Evergreen Needleleaf Forests            0.5618497
## Water Bodies-Evergreen Needleleaf Forests                        3.6247542
## Woody Savannas-Evergreen Needleleaf Forests                     -4.2238485
## Open Shrublands-Grasslands                                       3.0648412
## Permanent Snow and Ice-Grasslands                                8.8690548
## Permanent Wetlands-Grasslands                                    7.6839290
## Savannas-Grasslands                                              3.0763126
## Urban and Built-up Lands-Grasslands                              8.8625635
## Water Bodies-Grasslands                                         11.9955553
## Woody Savannas-Grasslands                                        3.6943593
## Permanent Snow and Ice-Open Shrublands                           8.9526641
## Permanent Wetlands-Open Shrublands                               7.8156830
## Savannas-Open Shrublands                                         4.2127560
## Urban and Built-up Lands-Open Shrublands                         9.0882536
## Water Bodies-Open Shrublands                                    12.0190875
## Woody Savannas-Open Shrublands                                   4.8308027
## Permanent Wetlands-Permanent Snow and Ice                       17.8823395
## Savannas-Permanent Snow and Ice                                 18.3803995
## Urban and Built-up Lands-Permanent Snow and Ice                 20.1027368
## Water Bodies-Permanent Snow and Ice                             20.6117186
## Woody Savannas-Permanent Snow and Ice                           18.9984463
## Savannas-Permanent Wetlands                                     15.5603029
## Urban and Built-up Lands-Permanent Wetlands                     17.6235617
## Water Bodies-Permanent Wetlands                                 18.6120929
## Woody Savannas-Permanent Wetlands                               16.1783496
## Urban and Built-up Lands-Savannas                                7.9992683
## Water Bodies-Savannas                                           11.0449480
## Woody Savannas-Savannas                                          3.2936578
## Water Bodies-Urban and Built-up Lands                           13.5426477
## Woody Savannas-Urban and Built-up Lands                          9.8551968
## Woody Savannas-Water Bodies                                     23.4171988
##                                                                      p adj
## Closed Shrublands-Barren                                        0.9999991
## Cropland/Natural Vegetation Mosaics-Barren                      0.9941217
## Croplands-Barren                                                0.1076959
## Evergreen Broadleaf Forests-Barren                              0.1742792
## Evergreen Needleleaf Forests-Barren                             0.0000000
## Grasslands-Barren                                               1.0000000
## Open Shrublands-Barren                                          1.0000000
## Permanent Snow and Ice-Barren                                   0.9982224
## Permanent Wetlands-Barren                                       0.9989455
## Savannas-Barren                                                 0.9997072
## Urban and Built-up Lands-Barren                                 1.0000000
## Water Bodies-Barren                                             0.9994202
## Woody Savannas-Barren                                           0.9507990
## Cropland/Natural Vegetation Mosaics-Closed Shrublands           0.9992046
## Croplands-Closed Shrublands                                     0.6406651
## Evergreen Broadleaf Forests-Closed Shrublands                   0.2847335
## Evergreen Needleleaf Forests-Closed Shrublands                  0.0000000
## Grasslands-Closed Shrublands                                    0.9996217
```

```
## Open Shrublands-Closed Shrublands                                    0.9999997
## Permanent Snow and Ice-Closed Shrublands                             0.9934846
## Permanent Wetlands-Closed Shrublands                                 0.9947294
## Savannas-Closed Shrublands                                           1.0000000
## Urban and Built-up Lands-Closed Shrublands                           1.0000000
## Water Bodies-Closed Shrublands                                       0.9980517
## Woody Savannas-Closed Shrublands                                     0.9998138
## Croplands-Cropland/Natural Vegetation Mosaics                        1.0000000
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics      0.9910784
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics     0.9163742
## Grasslands-Cropland/Natural Vegetation Mosaics                       0.9877338
## Open Shrublands-Cropland/Natural Vegetation Mosaics                  0.9948149
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics           0.9307640
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics               0.9334811
## Savannas-Cropland/Natural Vegetation Mosaics                         0.9996282
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics         0.9998336
## Water Bodies-Cropland/Natural Vegetation Mosaics                     0.9690714
## Woody Savannas-Cropland/Natural Vegetation Mosaics                   0.9999816
## Evergreen Broadleaf Forests-Croplands                               0.6584278
## Evergreen Needleleaf Forests-Croplands                              0.0000000
## Grasslands-Croplands                                                0.0000000
## Open Shrublands-Croplands                                           0.2161474
## Permanent Snow and Ice-Croplands                                    0.9008266
## Permanent Wetlands-Croplands                                        0.8728463
## Savannas-Croplands                                                  0.3535693
## Urban and Built-up Lands-Croplands                                  0.9998656
## Water Bodies-Croplands                                              0.9710058
## Woody Savannas-Croplands                                            0.9573149
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests            1.0000000
## Grasslands-Evergreen Broadleaf Forests                              0.1201124
## Open Shrublands-Evergreen Broadleaf Forests                         0.1853504
## Permanent Snow and Ice-Evergreen Broadleaf Forests                  0.2827238
## Permanent Wetlands-Evergreen Broadleaf Forests                      0.2287107
## Savannas-Evergreen Broadleaf Forests                                0.3065515
## Urban and Built-up Lands-Evergreen Broadleaf Forests                0.6255812
## Water Bodies-Evergreen Broadleaf Forests                            0.5044215
## Woody Savannas-Evergreen Broadleaf Forests                          0.4479887
## Grasslands-Evergreen Needleleaf Forests                             0.0000000
## Open Shrublands-Evergreen Needleleaf Forests                        0.0000000
## Permanent Snow and Ice-Evergreen Needleleaf Forests                 0.0747357
## Permanent Wetlands-Evergreen Needleleaf Forests                     0.0263733
## Savannas-Evergreen Needleleaf Forests                               0.0000000
## Urban and Built-up Lands-Evergreen Needleleaf Forests               0.0959614
## Water Bodies-Evergreen Needleleaf Forests                           0.3175277
## Woody Savannas-Evergreen Needleleaf Forests                         0.0000000
## Open Shrublands-Grasslands                                          1.0000000
## Permanent Snow and Ice-Grasslands                                   0.9985828
## Permanent Wetlands-Grasslands                                       0.9991883
## Savannas-Grasslands                                                 0.9116307
## Urban and Built-up Lands-Grasslands                                 1.0000000
## Water Bodies-Grasslands                                             0.9995499
## Woody Savannas-Grasslands                                           0.2609425
## Permanent Snow and Ice-Open Shrublands                              0.9981990
## Permanent Wetlands-Open Shrublands                                  0.9989337
```

```
## Savannas-Open Shrublands                                        0.9998867
## Urban and Built-up Lands-Open Shrublands                        1.0000000
## Water Bodies-Open Shrublands                                    0.9994073
## Woody Savannas-Open Shrublands                                  0.9730800
## Permanent Wetlands-Permanent Snow and Ice                       1.0000000
## Savannas-Permanent Snow and Ice                                 0.9886404
## Urban and Built-up Lands-Permanent Snow and Ice                 0.9993664
## Water Bodies-Permanent Snow and Ice                             1.0000000
## Woody Savannas-Permanent Snow and Ice                           0.9699765
## Savannas-Permanent Wetlands                                     0.9894996
## Urban and Built-up Lands-Permanent Wetlands                     0.9997507
## Water Bodies-Permanent Wetlands                                 1.0000000
## Woody Savannas-Permanent Wetlands                               0.9670995
## Urban and Built-up Lands-Savannas                               1.0000000
## Water Bodies-Savannas                                           0.9967758
## Woody Savannas-Savannas                                         0.9999612
## Water Bodies-Urban and Built-up Lands                           0.9996650
## Woody Savannas-Urban and Built-up Lands                         0.9999999
## Woody Savannas-Water Bodies                                     0.9915641
```

```r
# There is a significant difference in fast fire event length between september and january
#(january is higher, p=0.0000328). There is a significant difference in fast fire event length
#between March and February (February is higher, p= 0.0439883). There is a significant difference
#in fast fire event length between September and February (February is higher, p= 0.0000000).
#Other months between which there are significant differences in event length are: april-march,
#may-march, november-march, july-april, august-april, september-april, july-may,august-may,
#september-may, october-may, july-june, september-june, september-july, november-july,
#september-august, november-august, october-september, november-september, december-september,
#november-october. It seems like september is consistently significantly different from the event
#lengths in other months.

# There is a significant difference in event length between: Evergreen Needleleaf Forests-Closed
#Shrublands(*), Evergreen Broadleaf Forests-Croplands(*), Evergreen Needleleaf Forests-Croplands(*),
#Grasslands-Croplands(*), Grasslands(*)-Evergreen Needleleaf Forests, Open Shrublands(*)-Evergreen
#Needleleaf Forests, Permanent Wetlands(*)-Evergreen Needleleaf Forests,
#Savannas(*)-Evergreen Needleleaf Forests, Woody Savannas(*)-Evergreen Needleleaf Forests.
#(*) is next to the one that is higher.

# Format as lm
eventlength_aov2 <- lm(data = event_length, event_dur ~ ig_month + lc_name)
summary(eventlength_aov2)
```

```
##
## Call:
## lm(formula = event_dur ~ ig_month + lc_name, data = event_length)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.445  -4.964  -2.337   2.895  83.542
##
## Coefficients:
##                                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)                            5.17532    1.06826   4.845 1.31e-06
## ig_month2                             -0.88031    0.97518  -0.903 0.366729
```

```
## ig_month3                                    2.12217    1.01593   2.089 0.036776
## ig_month4                                   -1.32381    0.97018  -1.364 0.172483
## ig_month5                                   -0.64565    0.87878  -0.735 0.462554
## ig_month6                                    0.35559    0.85318   0.417 0.676855
## ig_month7                                    1.86012    0.85421   2.178 0.029492
## ig_month8                                    0.98085    0.85846   1.143 0.253283
## ig_month9                                    3.84740    0.83360   4.615 4.04e-06
## ig_month10                                  -0.16709    0.84577  -0.198 0.843397
## ig_month11                                  -2.35944    0.84773  -2.783 0.005406
## ig_month12                                  -1.43265    1.10957  -1.291 0.196712
## lc_nameClosed Shrublands                     0.72175    1.10428   0.654 0.513406
## lc_nameCropland/Natural Vegetation Mosaics   4.15297    2.86593   1.449 0.147387
## lc_nameCroplands                             2.80735    0.80878   3.471 0.000523
## lc_nameEvergreen Broadleaf Forests           8.52867    2.99527   2.847 0.004429
## lc_nameEvergreen Needleleaf Forests          8.42220    0.91756   9.179  < 2e-16
## lc_nameGrasslands                           -0.30526    0.79695  -0.383 0.701709
## lc_nameOpen Shrublands                      -0.06816    1.12411  -0.061 0.951655
## lc_namePermanent Snow and Ice               -4.55868    3.95655  -1.152 0.249309
## lc_namePermanent Wetlands                   -3.61694    3.36747  -1.074 0.282847
## lc_nameSavannas                              1.02339    0.95119   1.076 0.282032
## lc_nameUrban and Built-up Lands              0.27575    2.62873   0.105 0.916460
## lc_nameWater Bodies                         -4.62468    5.07154  -0.912 0.361878
## lc_nameWoody Savannas                        1.67732    0.94811   1.769 0.076943
##
## (Intercept)                                 ***
## ig_month2
## ig_month3                                   *
## ig_month4
## ig_month5
## ig_month6
## ig_month7                                   *
## ig_month8
## ig_month9                                   ***
## ig_month10
## ig_month11                                  **
## ig_month12
## lc_nameClosed Shrublands
## lc_nameCropland/Natural Vegetation Mosaics
## lc_nameCroplands                            ***
## lc_nameEvergreen Broadleaf Forests          **
## lc_nameEvergreen Needleleaf Forests         ***
## lc_nameGrasslands
## lc_nameOpen Shrublands
## lc_namePermanent Snow and Ice
## lc_namePermanent Wetlands
## lc_nameSavannas
## lc_nameUrban and Built-up Lands
## lc_nameWater Bodies
## lc_nameWoody Savannas                       .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.68 on 4304 degrees of freedom
## Multiple R-squared:  0.1053, Adjusted R-squared:  0.1004
```

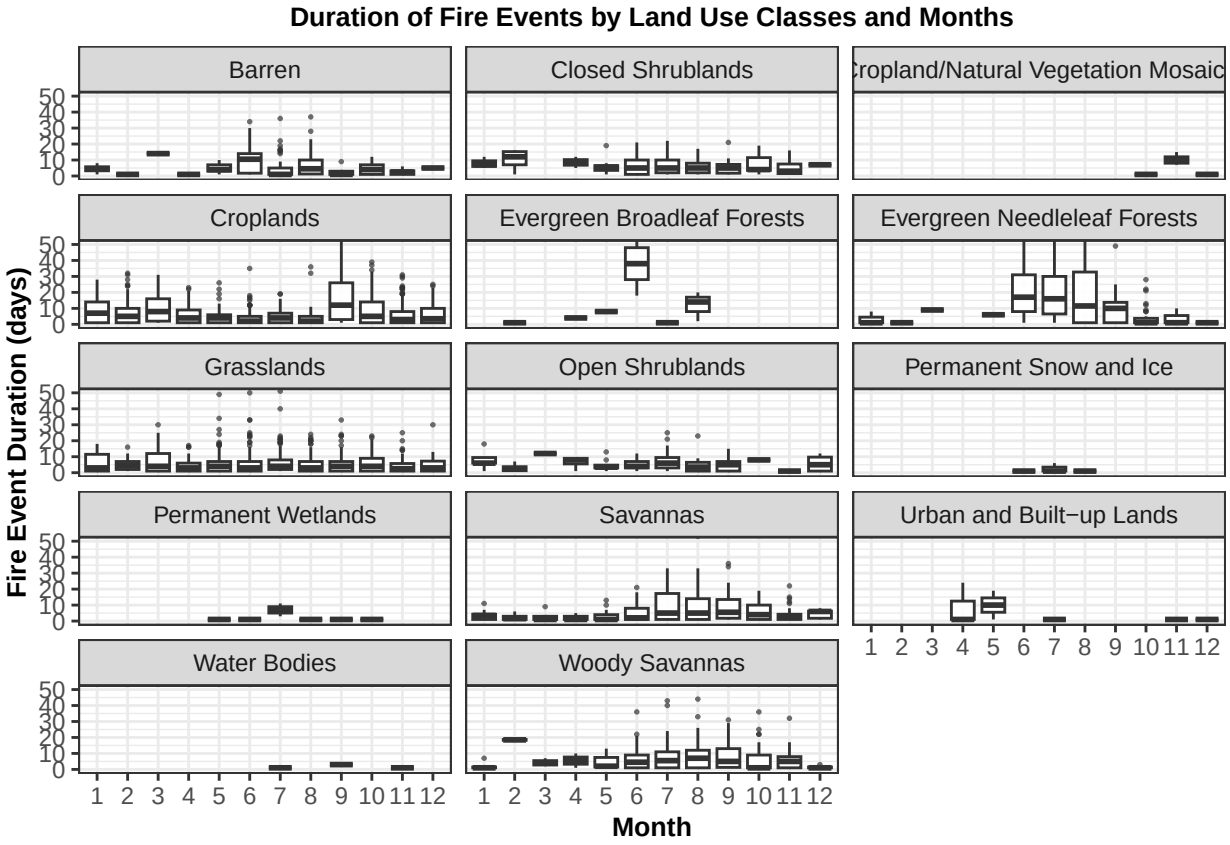
```
## F-statistic: 21.12 on 24 and 4304 DF,  p-value: < 2.2e-16
```

```
#The Multiple R-squared is 0.1053. The Adjusted R-squared is 0.1004. This means that these variables
#don't explain much of the variation, and more variables should be added to the model.
```

Figures

```
event_dur_fig <- ggplot(event_length,
                        aes(y = event_dur,
                            x = ig_month,
                            #color = lc_name
                            )
                        )+
  geom_boxplot(alpha = 0.9,
               size= 0.5,
               outlier.size = 0.3,
               outlier.alpha = 0.7)+
  labs(title = "Duration of Fire Events by Land Use Classes and Months",
       x= "Month",
       y="Fire Event Duration (days)"
       )+
  facet_wrap(vars(lc_name), ncol = 3)+
  coord_cartesian(ylim = c(0, 50))
  #theme(legend.position = "bottom", legend.size = 0.3)

event_dur_fig
```



The question this figure addresses is: What is the relationship between time of year and which land cover type sees the longest fast fires? One visible pattern is that Grasslands seems to have fires of relatively consistent length throughout the year. Evergeen Needleleaf Forests, on the other hand, generally have longer fires June-September than they do during the rest of the year. The x-axis units are in months of the year, and the y-axis units are in days. A limitation of this figure is that it merges 20 years of data, and so patterns over time are not visible here. Another limitation is that it doesn't clearly show the difference in number of events between these land cover types. The Summary Table below shows that the number of events ranges from 3 in Water Bodies to 1,670 in Croplands. Provisional findings: Some of the land cover types may not have enough data to include in our analysis. This includes Water Bodies, Permanent Snow and Ice, and Permanent Wetlands. Another finding is that Evergeen Needleleaf Forest appears to have longer fires in June-September than it does throughout the rest of the year. A question that comes up here is why there is no deciduous forest landcover type in this dataset.

Summary Table

```
# Group data
duration_table <- california_events_processed %>%
  group_by(lc_name) %>%
  summarize("Number of Fires" = n(),
            "Max Duration (days)" = max(event_dur),
            "Year (max dur)" = ig_year[which.max(event_dur)],
            "Max Size (km2)" = max(tot_ar_km2),
            "Year (max size)" = ig_year[which.max(tot_ar_km2)])%>%
           # "Mean km2/day" = mean(mx_grw_km2))%>%
  rename("Land Cover Type" = lc_name) %>%
  ungroup()

# formatting summary table
summary_duration <- gt(duration_table) %>%
  fmt_number(decimals = 0)%>% # rounding to a whole number
  tab_header(
    title = "Fire Duration and Size by Land Cover Type",
    subtitle = "California, 2000-2021")%>%
    tab_options(
      heading.title.font.size = "10pt",
      heading.subtitle.font.size = "8pt",
      table.font.size = "8pt",
      latex.use_longtable = FALSE )

summary_duration
```

Central Question ANOVA

###What is the relationship between land cover type and fire growth rate?

##Response variables: max, mean and min daily growth per event ##Predictor variable: Land cover

#We felt an ANOVA was the mosst appropriate to address this question because the response variables are continuous, the predictor is categorical, and individual fire events can be used as observational units.

Fire Duration and Size by Land Cover Type

California, 2000-2021

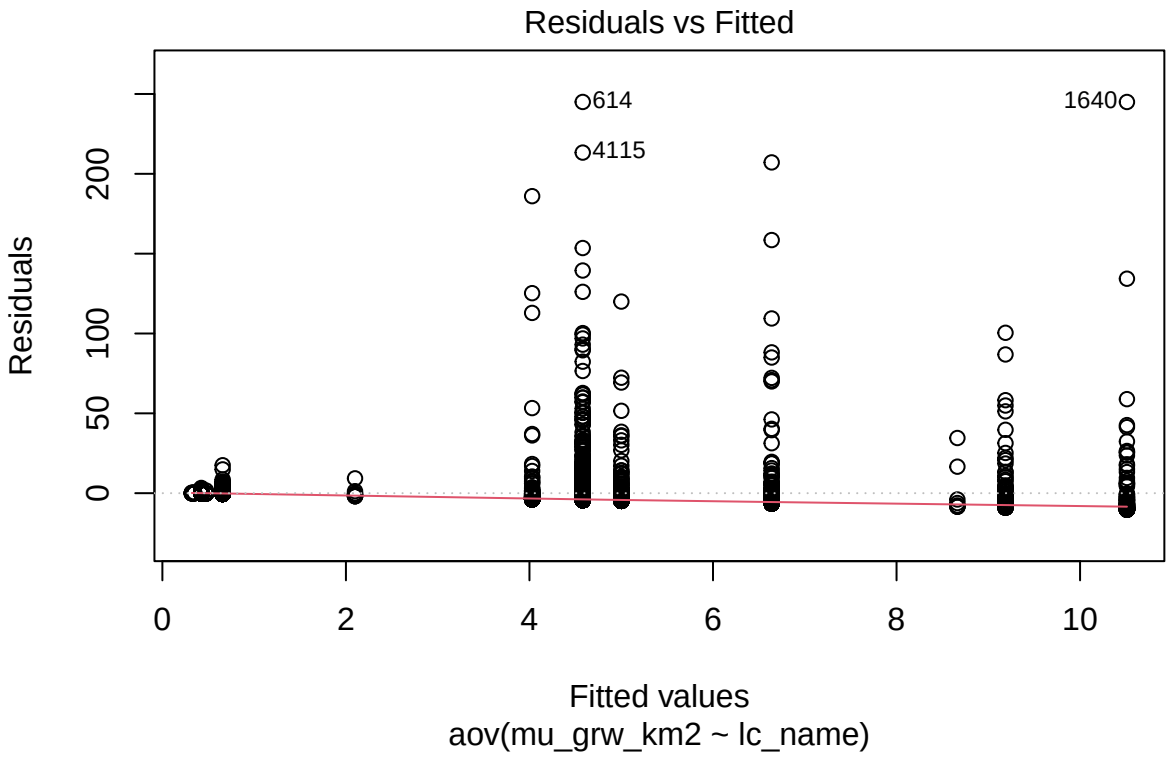
Land Cover Type	Number of Fires	Max Duration (days)	Year (max dur)	Max Size (km2)	Year (max size)
Barren	132	37	2005	32	2005
Closed Shrublands	118	22	2006	819	2007
Cropland/Natural Vegetation Mosaics	10	15	2018	39	2020
Croplands	1,670	53	2005	58	2010
Evergreen Broadleaf Forests	9	58	2008	875	2008
Evergreen Needleleaf Forests	285	99	2006	3,778	2020
Grasslands	1,494	59	2015	1,132	2020
Open Shrublands	110	25	2006	334	2006
Permanent Snow and Ice	5	6	2018	4	2018
Permanent Wetlands	7	11	2001	1	2018
Savannas	237	55	2007	1,450	2020
Urban and Built-up Lands	12	24	2009	3	2009
Water Bodies	3	3	2020	1	2020
Woody Savannas	237	59	2016	1,294	2020

```
#Load dplyr and make it so that there is only one row per fire event
library(dplyr)
fire_events <- california_events_processed |>
  distinct(id, .keep_all = TRUE)
#Make anova models for each with land cover type as the predictor variable
max_model <- aov(mx_grw_km2 ~ lc_name, data = fire_events)
min_model <- aov(mn_grw_km2 ~ lc_name, data = fire_events)
mean_model <- aov(mu_grw_km2 ~ lc_name, data = fire_events)

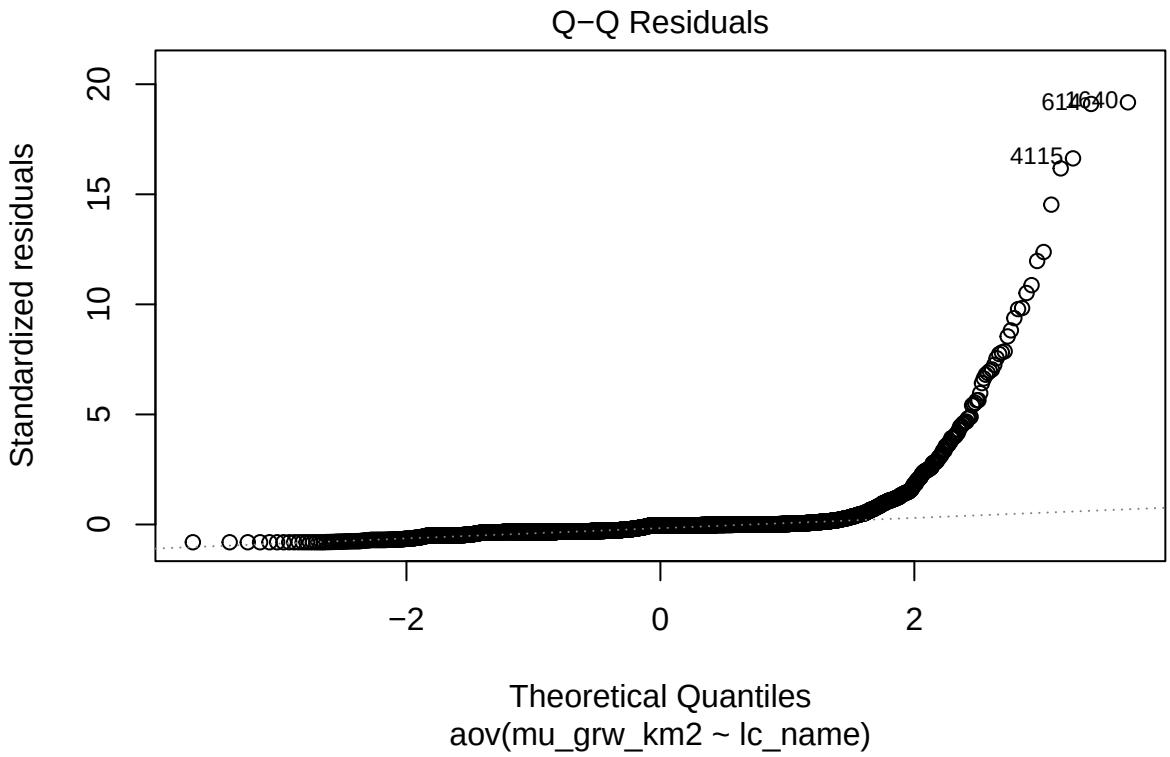
anova_summary <- data.frame(Metric = c("Maximum", "Mean", "Minimum"),
  F_value = c(summary(max_model)[[1]]["lc_name", "F value"],
    summary(mean_model)[[1]]["lc_name", "F value"],
    summary(min_model)[[1]]["lc_name", "F value"]),
  p_value = c(summary(max_model)[[1]]["lc_name", "Pr(>F)"],
    summary(mean_model)[[1]]["lc_name", "Pr(>F)"],
    summary(min_model)[[1]]["lc_name", "Pr(>F)"]))
anova_summary
```

```
##    Metric   F_value      p_value
## 1 Maximum 11.556221 5.552504e-25
## 2    Mean 13.855526 8.035791e-31
## 3 Minimum  7.498301 7.826727e-15
```

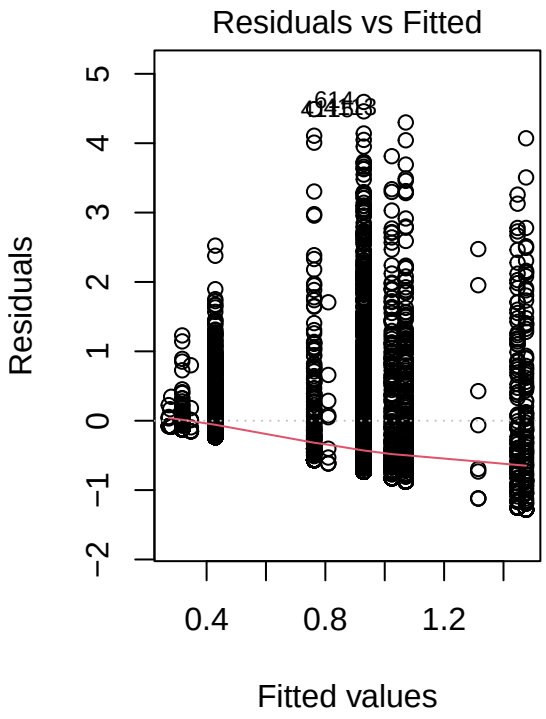
```
#plot residuals vs. fitted values for the mean model as a diagnostic
plot(mean_model, which = 1)
```

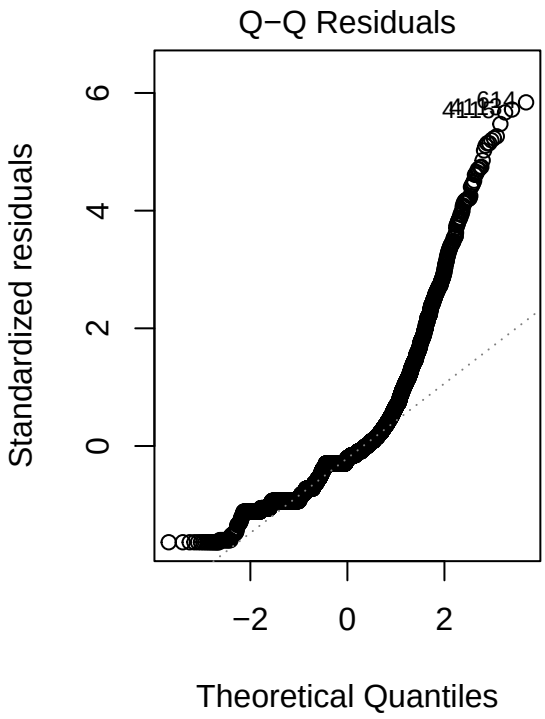



```
#plot Q-Q plot as a diagnostic
plot(mean_model, which = 2)
```




```r
#Both the Q-Q and the residual plots raise red flags. Maybe a log transform will help?
mean_model_log <- aov(log1p(mu_grw_km2) ~ lc_name, data = fire_events)
#Diagnostic test again
par(mfrow = c(1, 2))
plot(mean_model_log, which = 1)
plot(mean_model_log, which = 2)
```






```
#Red flags are still there with the log transforms

#Since Pr(>F)<.001 for each anova model, we will run a post hoc Tukey test to interpret
#this statistical difference
TukeyHSD(max_model)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = mx_grw_km2 ~ lc_name, data = fire_events)
##
## $lc_name
##                                                          diff
## Closed Shrublands-Barren                          14.014857508
## Cropland/Natural Vegetation Mosaics-Barren         2.466948541
## Croplands-Barren                                   0.382188647
## Evergreen Broadleaf Forests-Barren                13.414540879
## Evergreen Needleleaf Forests-Barren               12.312626356
## Grasslands-Barren                                  5.936689037
## Open Shrublands-Barren                            11.197702217
## Permanent Snow and Ice-Barren                     -0.108955539
## Permanent Wetlands-Barren                         -0.200952113
## Savannas-Barren                                    5.688640106
## Urban and Built-up Lands-Barren                   -0.001626202
## Water Bodies-Barren                               -0.180508430
## Woody Savannas-Barren                              7.565318465
```

```
## Cropland/Natural Vegetation Mosaics-Closed Shrublands            -11.547908967
## Croplands-Closed Shrublands                                      -13.632668860
## Evergreen Broadleaf Forests-Closed Shrublands                     -0.600316629
## Evergreen Needleleaf Forests-Closed Shrublands                    -1.702231152
## Grasslands-Closed Shrublands                                      -8.078168471
## Open Shrublands-Closed Shrublands                                 -2.817155291
## Permanent Snow and Ice-Closed Shrublands                         -14.123813046
## Permanent Wetlands-Closed Shrublands                             -14.215809621
## Savannas-Closed Shrublands                                        -8.326217402
## Urban and Built-up Lands-Closed Shrublands                       -14.016483710
## Water Bodies-Closed Shrublands                                   -14.195365938
## Woody Savannas-Closed Shrublands                                  -6.449539043
## Croplands-Cropland/Natural Vegetation Mosaics                     -2.084759894
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics   10.947592338
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics   9.845677815
## Grasslands-Cropland/Natural Vegetation Mosaics                     3.469740496
## Open Shrublands-Cropland/Natural Vegetation Mosaics                8.730753676
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics        -2.575904080
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics            -2.667900654
## Savannas-Cropland/Natural Vegetation Mosaics                       3.221691565
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics      -2.468574743
## Water Bodies-Cropland/Natural Vegetation Mosaics                  -2.647456971
## Woody Savannas-Cropland/Natural Vegetation Mosaics                 5.098369924
## Evergreen Broadleaf Forests-Croplands                             13.032352232
## Evergreen Needleleaf Forests-Croplands                            11.930437709
## Grasslands-Croplands                                               5.554500390
## Open Shrublands-Croplands                                         10.815513569
## Permanent Snow and Ice-Croplands                                  -0.491144186
## Permanent Wetlands-Croplands                                      -0.583140760
## Savannas-Croplands                                                 5.306451459
## Urban and Built-up Lands-Croplands                                -0.383814849
## Water Bodies-Croplands                                            -0.562697077
## Woody Savannas-Croplands                                           7.183129818
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests          -1.101914523
## Grasslands-Evergreen Broadleaf Forests                            -7.477851842
## Open Shrublands-Evergreen Broadleaf Forests                       -2.216838662
## Permanent Snow and Ice-Evergreen Broadleaf Forests               -13.523496418
## Permanent Wetlands-Evergreen Broadleaf Forests                   -13.615492992
## Savannas-Evergreen Broadleaf Forests                              -7.725900773
## Urban and Built-up Lands-Evergreen Broadleaf Forests             -13.416167081
## Water Bodies-Evergreen Broadleaf Forests                         -13.595049309
## Woody Savannas-Evergreen Broadleaf Forests                        -5.849222414
## Grasslands-Evergreen Needleleaf Forests                           -6.375937319
## Open Shrublands-Evergreen Needleleaf Forests                      -1.114924139
## Permanent Snow and Ice-Evergreen Needleleaf Forests              -12.421581895
## Permanent Wetlands-Evergreen Needleleaf Forests                  -12.513578469
## Savannas-Evergreen Needleleaf Forests                             -6.623986250
## Urban and Built-up Lands-Evergreen Needleleaf Forests            -12.314252558
## Water Bodies-Evergreen Needleleaf Forests                        -12.493134786
## Woody Savannas-Evergreen Needleleaf Forests                       -4.747307891
## Open Shrublands-Grasslands                                         5.261013179
## Permanent Snow and Ice-Grasslands                                 -6.045644576
## Permanent Wetlands-Grasslands                                     -6.137641150
## Savannas-Grasslands                                               -0.248048931
```

```
## Urban and Built-up Lands-Grasslands                                  -5.938315239
## Water Bodies-Grasslands                                              -6.117197467
## Woody Savannas-Grasslands                                             1.628629428
## Permanent Snow and Ice-Open Shrublands                              -11.306657755
## Permanent Wetlands-Open Shrublands                                  -11.398654330
## Savannas-Open Shrublands                                             -5.509062111
## Urban and Built-up Lands-Open Shrublands                            -11.199328419
## Water Bodies-Open Shrublands                                        -11.378210646
## Woody Savannas-Open Shrublands                                       -3.632383752
## Permanent Wetlands-Permanent Snow and Ice                            -0.091996574
## Savannas-Permanent Snow and Ice                                       5.797595645
## Urban and Built-up Lands-Permanent Snow and Ice                       0.107329337
## Water Bodies-Permanent Snow and Ice                                  -0.071552891
## Woody Savannas-Permanent Snow and Ice                                 7.674274004
## Savannas-Permanent Wetlands                                           5.889592219
## Urban and Built-up Lands-Permanent Wetlands                           0.199325911
## Water Bodies-Permanent Wetlands                                       0.020443683
## Woody Savannas-Permanent Wetlands                                     7.766270578
## Urban and Built-up Lands-Savannas                                    -5.690266308
## Water Bodies-Savannas                                                -5.869148536
## Woody Savannas-Savannas                                               1.876678359
## Water Bodies-Urban and Built-up Lands                                -0.178882228
## Woody Savannas-Urban and Built-up Lands                               7.566944667
## Woody Savannas-Water Bodies                                           7.745826895
##                                                                             lwr
## Closed Shrublands-Barren                                              4.7760070
## Cropland/Natural Vegetation Mosaics-Barren                          -21.4515058
## Croplands-Barren                                                     -6.2111912
## Evergreen Broadleaf Forests-Barren                                  -11.7087913
## Evergreen Needleleaf Forests-Barren                                   4.6348663
## Grasslands-Barren                                                    -0.6850782
## Open Shrublands-Barren                                                1.7831368
## Permanent Snow and Ice-Barren                                       -33.3338967
## Permanent Wetlands-Barren                                           -28.4853748
## Savannas-Barren                                                      -2.2314104
## Urban and Built-up Lands-Barren                                     -21.9893140
## Water Bodies-Barren                                                 -42.7594828
## Woody Savannas-Barren                                                -0.3547320
## Cropland/Natural Vegetation Mosaics-Closed Shrublands               -35.5660774
## Croplands-Closed Shrublands                                         -20.5790790
## Evergreen Broadleaf Forests-Closed Shrublands                       -25.8185992
## Evergreen Needleleaf Forests-Closed Shrublands                       -9.6852103
## Grasslands-Closed Shrublands                                        -15.0515290
## Open Shrublands-Closed Shrublands                                   -12.4822471
## Permanent Snow and Ice-Closed Shrublands                            -47.4206097
## Permanent Wetlands-Closed Shrublands                                -42.5846045
## Savannas-Closed Shrublands                                          -16.5424912
## Urban and Built-up Lands-Closed Shrublands                          -36.1126004
## Water Bodies-Closed Shrublands                                      -56.8304338
## Woody Savannas-Closed Shrublands                                    -14.6658129
## Croplands-Cropland/Natural Vegetation Mosaics                       -25.2145829
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics     -22.5590917
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics    -13.6162926
## Grasslands-Cropland/Natural Vegetation Mosaics                      -19.6681906
```

```
## Open Shrublands-Cropland/Natural Vegetation Mosaics              -15.3555515
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics       -42.5185225
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics           -38.6056889
## Savannas-Cropland/Natural Vegetation Mosaics                     -20.3206799
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics     -33.6931559
## Water Bodies-Cropland/Natural Vegetation Mosaics                 -50.6525099
## Woody Savannas-Cropland/Natural Vegetation Mosaics               -18.4440015
## Evergreen Broadleaf Forests-Croplands                           -11.3413646
## Evergreen Needleleaf Forests-Croplands                            7.2566539
## Grasslands-Croplands                                              2.9575682
## Open Shrublands-Croplands                                         3.6370529
## Permanent Snow and Ice-Croplands                                -33.1529410
## Permanent Wetlands-Croplands                                    -28.2038725
## Savannas-Croplands                                                0.2444899
## Urban and Built-up Lands-Croplands                              -21.5109223
## Water Bodies-Croplands                                          -42.7037140
## Woody Savannas-Croplands                                          2.1211683
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests        -25.7910501
## Grasslands-Evergreen Broadleaf Forests                          -31.8592632
## Open Shrublands-Evergreen Broadleaf Forests                     -27.5000240
## Permanent Snow and Ice-Evergreen Broadleaf Forests              -54.1990681
## Permanent Wetlands-Evergreen Broadleaf Forests                  -50.3661934
## Savannas-Evergreen Broadleaf Forests                            -32.4914538
## Urban and Built-up Lands-Evergreen Broadleaf Forests            -45.5730300
## Water Bodies-Evergreen Broadleaf Forests                        -62.2116563
## Woody Savannas-Evergreen Broadleaf Forests                      -30.6147754
## Grasslands-Evergreen Needleleaf Forests                         -11.0896830
## Open Shrublands-Evergreen Needleleaf Forests                     -9.3006214
## Permanent Snow and Ice-Evergreen Needleleaf Forests             -45.3194285
## Permanent Wetlands-Evergreen Needleleaf Forests                 -40.4130438
## Savannas-Evergreen Needleleaf Forests                           -13.0348206
## Urban and Built-up Lands-Evergreen Needleleaf Forests           -33.8044832
## Water Bodies-Evergreen Needleleaf Forests                       -54.8173672
## Woody Savannas-Evergreen Needleleaf Forests                     -11.1581422
## Open Shrublands-Grasslands                                       -1.9435299
## Permanent Snow and Ice-Grasslands                               -38.7131837
## Permanent Wetlands-Grasslands                                   -33.7651630
## Savannas-Grasslands                                              -5.3469310
## Urban and Built-up Lands-Grasslands                             -27.0742991
## Water Bodies-Grasslands                                         -48.2626652
## Woody Savannas-Grasslands                                        -3.4702526
## Permanent Snow and Ice-Open Shrublands                          -44.6526373
## Permanent Wetlands-Open Shrublands                              -39.8251597
## Savannas-Open Shrublands                                        -13.9224348
## Urban and Built-up Lands-Open Shrublands                        -33.3694898
## Water Bodies-Open Shrublands                                    -54.0517000
## Woody Savannas-Open Shrublands                                  -12.0457564
## Permanent Wetlands-Permanent Snow and Ice                       -42.7924518
## Savannas-Permanent Snow and Ice                                 -27.1576393
## Urban and Built-up Lands-Permanent Snow and Ice                 -38.7099184
## Water Bodies-Permanent Snow and Ice                             -53.3283775
## Woody Savannas-Permanent Snow and Ice                           -25.2809609
## Savannas-Permanent Wetlands                                     -22.0775200
## Urban and Built-up Lands-Permanent Wetlands                     -34.4833871
```

```
## Water Bodies-Permanent Wetlands                                    -50.3025254
## Woody Savannas-Permanent Wetlands                                  -20.2008416
## Urban and Built-up Lands-Savannas                                  -27.2682461
## Water Bodies-Savannas                                              -48.2380033
## Woody Savannas-Savannas                                             -4.8224258
## Water Bodies-Urban and Built-up Lands                              -47.2517095
## Woody Savannas-Urban and Built-up Lands                            -14.0110351
## Woody Savannas-Water Bodies                                        -34.6230278
##                                                                            upr
## Closed Shrublands-Barren                                           23.2537080
## Cropland/Natural Vegetation Mosaics-Barren                         26.3854029
## Croplands-Barren                                                    6.9755685
## Evergreen Broadleaf Forests-Barren                                 38.5378730
## Evergreen Needleleaf Forests-Barren                                19.9903864
## Grasslands-Barren                                                  12.5584563
## Open Shrublands-Barren                                             20.6122677
## Permanent Snow and Ice-Barren                                      33.1159856
## Permanent Wetlands-Barren                                          28.0834706
## Savannas-Barren                                                    13.6086906
## Urban and Built-up Lands-Barren                                    21.9860616
## Water Bodies-Barren                                                42.3984659
## Woody Savannas-Barren                                              15.4853689
## Cropland/Natural Vegetation Mosaics-Closed Shrublands              12.4702594
## Croplands-Closed Shrublands                                        -6.6862587
## Evergreen Broadleaf Forests-Closed Shrublands                      24.6179659
## Evergreen Needleleaf Forests-Closed Shrublands                      6.2807480
## Grasslands-Closed Shrublands                                       -1.1048079
## Open Shrublands-Closed Shrublands                                   6.8479365
## Permanent Snow and Ice-Closed Shrublands                           19.1729836
## Permanent Wetlands-Closed Shrublands                               14.1529853
## Savannas-Closed Shrublands                                         -0.1099436
## Urban and Built-up Lands-Closed Shrublands                          8.0796329
## Water Bodies-Closed Shrublands                                     28.4397019
## Woody Savannas-Closed Shrublands                                    1.7667348
## Croplands-Cropland/Natural Vegetation Mosaics                      21.0450631
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics    44.4542764
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics   33.3076482
## Grasslands-Cropland/Natural Vegetation Mosaics                     26.6076716
## Open Shrublands-Cropland/Natural Vegetation Mosaics                32.8170589
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics         37.3667144
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics             33.2698876
## Savannas-Cropland/Natural Vegetation Mosaics                       26.7640630
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics       28.7560064
## Water Bodies-Cropland/Natural Vegetation Mosaics                   45.3575960
## Woody Savannas-Cropland/Natural Vegetation Mosaics                 28.6407414
## Evergreen Broadleaf Forests-Croplands                              37.4060691
## Evergreen Needleleaf Forests-Croplands                             16.6042215
## Grasslands-Croplands                                                8.1514326
## Open Shrublands-Croplands                                          17.9939743
## Permanent Snow and Ice-Croplands                                   32.1706526
## Permanent Wetlands-Croplands                                       27.0375910
## Savannas-Croplands                                                 10.3684130
## Urban and Built-up Lands-Croplands                                 20.7432926
## Water Bodies-Croplands                                             41.5783199
```

```
## Woody Savannas-Croplands                                          12.2450914
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests          23.5872211
## Grasslands-Evergreen Broadleaf Forests                            16.9035595
## Open Shrublands-Evergreen Broadleaf Forests                       23.0663466
## Permanent Snow and Ice-Evergreen Broadleaf Forests                27.1520752
## Permanent Wetlands-Evergreen Broadleaf Forests                    23.1352075
## Savannas-Evergreen Broadleaf Forests                              17.0396522
## Urban and Built-up Lands-Evergreen Broadleaf Forests              18.7406958
## Water Bodies-Evergreen Broadleaf Forests                          35.0215576
## Woody Savannas-Evergreen Broadleaf Forests                        18.9163306
## Grasslands-Evergreen Needleleaf Forests                           -1.6621916
## Open Shrublands-Evergreen Needleleaf Forests                       7.0707731
## Permanent Snow and Ice-Evergreen Needleleaf Forests               20.4762647
## Permanent Wetlands-Evergreen Needleleaf Forests                   15.3858868
## Savannas-Evergreen Needleleaf Forests                             -0.2131519
## Urban and Built-up Lands-Evergreen Needleleaf Forests              9.1759781
## Water Bodies-Evergreen Needleleaf Forests                         29.8310977
## Woody Savannas-Evergreen Needleleaf Forests                        1.6635264
## Open Shrublands-Grasslands                                        12.4655563
## Permanent Snow and Ice-Grasslands                                 26.6218945
## Permanent Wetlands-Grasslands                                     21.4898807
## Savannas-Grasslands                                                4.8508331
## Urban and Built-up Lands-Grasslands                               15.1976686
## Water Bodies-Grasslands                                           36.0282703
## Woody Savannas-Grasslands                                          6.7275115
## Permanent Snow and Ice-Open Shrublands                            22.0393218
## Permanent Wetlands-Open Shrublands                                17.0278510
## Savannas-Open Shrublands                                           2.9043106
## Urban and Built-up Lands-Open Shrublands                          10.9708329
## Water Bodies-Open Shrublands                                      31.2952787
## Woody Savannas-Open Shrublands                                     4.7809889
## Permanent Wetlands-Permanent Snow and Ice                         42.6084587
## Savannas-Permanent Snow and Ice                                   38.7528306
## Urban and Built-up Lands-Permanent Snow and Ice                   38.9245771
## Water Bodies-Permanent Snow and Ice                               53.1852717
## Woody Savannas-Permanent Snow and Ice                             40.6295089
## Savannas-Permanent Wetlands                                       33.8567044
## Urban and Built-up Lands-Permanent Wetlands                       34.8820389
## Water Bodies-Permanent Wetlands                                   50.3434128
## Woody Savannas-Permanent Wetlands                                 35.7333828
## Urban and Built-up Lands-Savannas                                 15.8877135
## Water Bodies-Savannas                                             36.4997062
## Woody Savannas-Savannas                                            8.5757825
## Water Bodies-Urban and Built-up Lands                             46.8939450
## Woody Savannas-Urban and Built-up Lands                           29.1449244
## Woody Savannas-Water Bodies                                       50.1146816
##                                                                        p adj
## Closed Shrublands-Barren                                          0.0000330
## Cropland/Natural Vegetation Mosaics-Barren                        1.0000000
## Croplands-Barren                                                  1.0000000
## Evergreen Broadleaf Forests-Barren                                0.8802527
## Evergreen Needleleaf Forests-Barren                               0.0000070
## Grasslands-Barren                                                 0.1354219
## Open Shrublands-Barren                                            0.0051895
```

```
## Permanent Snow and Ice-Barren                                          1.0000000
## Permanent Wetlands-Barren                                              1.0000000
## Savannas-Barren                                                        0.4728003
## Urban and Built-up Lands-Barren                                        1.0000000
## Water Bodies-Barren                                                    1.0000000
## Woody Savannas-Barren                                                  0.0787226
## Cropland/Natural Vegetation Mosaics-Closed Shrublands                  0.9430055
## Croplands-Closed Shrublands                                            0.0000000
## Evergreen Broadleaf Forests-Closed Shrublands                          1.0000000
## Evergreen Needleleaf Forests-Closed Shrublands                         0.9999847
## Grasslands-Closed Shrublands                                           0.0077911
## Open Shrublands-Closed Shrublands                                      0.9994754
## Permanent Snow and Ice-Closed Shrublands                               0.9793363
## Permanent Wetlands-Closed Shrublands                                   0.9226563
## Savannas-Closed Shrublands                                             0.0433799
## Urban and Built-up Lands-Closed Shrublands                             0.6814424
## Water Bodies-Closed Shrublands                                         0.9978887
## Woody Savannas-Closed Shrublands                                       0.3187874
## Croplands-Cropland/Natural Vegetation Mosaics                          1.0000000
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics        0.9982580
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics       0.9811693
## Grasslands-Cropland/Natural Vegetation Mosaics                         0.9999998
## Open Shrublands-Cropland/Natural Vegetation Mosaics                    0.9950992
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics             1.0000000
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics                 1.0000000
## Savannas-Cropland/Natural Vegetation Mosaics                           0.9999999
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics           1.0000000
## Water Bodies-Cropland/Natural Vegetation Mosaics                       1.0000000
## Woody Savannas-Cropland/Natural Vegetation Mosaics                     0.9999817
## Evergreen Broadleaf Forests-Croplands                                  0.8791564
## Evergreen Needleleaf Forests-Croplands                                 0.0000000
## Grasslands-Croplands                                                   0.0000000
## Open Shrublands-Croplands                                              0.0000395
## Permanent Snow and Ice-Croplands                                       1.0000000
## Permanent Wetlands-Croplands                                           1.0000000
## Savannas-Croplands                                                     0.0295445
## Urban and Built-up Lands-Croplands                                     1.0000000
## Water Bodies-Croplands                                                 1.0000000
## Woody Savannas-Croplands                                               0.0001719
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests               1.0000000
## Grasslands-Evergreen Broadleaf Forests                                 0.9990974
## Open Shrublands-Evergreen Broadleaf Forests                            1.0000000
## Permanent Snow and Ice-Evergreen Broadleaf Forests                     0.9979193
## Permanent Wetlands-Evergreen Broadleaf Forests                         0.9939559
## Savannas-Evergreen Broadleaf Forests                                   0.9989209
## Urban and Built-up Lands-Evergreen Broadleaf Forests                   0.9820989
## Water Bodies-Evergreen Broadleaf Forests                               0.9996655
## Woody Savannas-Evergreen Broadleaf Forests                             0.9999498
## Grasslands-Evergreen Needleleaf Forests                                0.0004933
## Open Shrublands-Evergreen Needleleaf Forests                           0.9999999
## Permanent Snow and Ice-Evergreen Needleleaf Forests                    0.9927655
## Permanent Wetlands-Evergreen Needleleaf Forests                        0.9670095
## Savannas-Evergreen Needleleaf Forests                                  0.0349465
## Urban and Built-up Lands-Evergreen Needleleaf Forests                  0.8142378
```

```
## Water Bodies-Evergreen Needleleaf Forests                          0.9993993
## Woody Savannas-Evergreen Needleleaf Forests                        0.4187721
## Open Shrublands-Grasslands                                         0.4434731
## Permanent Snow and Ice-Grasslands                                  0.9999972
## Permanent Wetlands-Grasslands                                      0.9999753
## Savannas-Grasslands                                                1.0000000
## Urban and Built-up Lands-Grasslands                                0.9996479
## Water Bodies-Grasslands                                            0.9999999
## Woody Savannas-Grasslands                                          0.9986205
## Permanent Snow and Ice-Open Shrublands                             0.9974625
## Permanent Wetlands-Open Shrublands                                 0.9874075
## Savannas-Open Shrublands                                           0.6317037
## Urban and Built-up Lands-Open Shrublands                           0.9180778
## Water Bodies-Open Shrublands                                       0.9998021
## Woody Savannas-Open Shrublands                                     0.9759744
## Permanent Wetlands-Permanent Snow and Ice                          1.0000000
## Savannas-Permanent Snow and Ice                                    0.9999985
## Urban and Built-up Lands-Permanent Snow and Ice                    1.0000000
## Water Bodies-Permanent Snow and Ice                                1.0000000
## Woody Savannas-Permanent Snow and Ice                              0.9999574
## Savannas-Permanent Wetlands                                        0.9999868
## Urban and Built-up Lands-Permanent Wetlands                        1.0000000
## Water Bodies-Permanent Wetlands                                    1.0000000
## Woody Savannas-Permanent Wetlands                                  0.9996901
## Urban and Built-up Lands-Savannas                                  0.9998250
## Water Bodies-Savannas                                              0.9999999
## Woody Savannas-Savannas                                            0.9996589
## Water Bodies-Urban and Built-up Lands                              1.0000000
## Woody Savannas-Urban and Built-up Lands                            0.9964524
## Woody Savannas-Water Bodies                                        0.9999976
```

```
TukeyHSD(mean_model)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = mu_grw_km2 ~ lc_name, data = fire_events)
##
## $lc_name
##                                                                    diff
## Closed Shrublands-Barren                                     10.086806154
## Cropland/Natural Vegetation Mosaics-Barren                    1.675170787
## Croplands-Barren                                              0.232847810
## Evergreen Broadleaf Forests-Barren                            8.239098407
## Evergreen Needleleaf Forests-Barren                           6.215522208
## Grasslands-Barren                                             4.156394695
## Open Shrublands-Barren                                        8.763363407
## Permanent Snow and Ice-Barren                                -0.078313487
## Permanent Wetlands-Barren                                    -0.103061103
## Savannas-Barren                                               3.605153154
## Urban and Built-up Lands-Barren                               0.051501660
## Water Bodies-Barren                                          -0.094883630
## Woody Savannas-Barren                                         4.577268265
## Cropland/Natural Vegetation Mosaics-Closed Shrublands        -8.411635367
```

```
## Croplands-Closed Shrublands                                          -9.853958344
## Evergreen Broadleaf Forests-Closed Shrublands                        -1.847707747
## Evergreen Needleleaf Forests-Closed Shrublands                       -3.871283946
## Grasslands-Closed Shrublands                                         -5.930411458
## Open Shrublands-Closed Shrublands                                    -1.323442746
## Permanent Snow and Ice-Closed Shrublands                            -10.165119640
## Permanent Wetlands-Closed Shrublands                                -10.189867257
## Savannas-Closed Shrublands                                           -6.481652999
## Urban and Built-up Lands-Closed Shrublands                          -10.035304494
## Water Bodies-Closed Shrublands                                      -10.181689783
## Woody Savannas-Closed Shrublands                                     -5.509537889
## Croplands-Cropland/Natural Vegetation Mosaics                        -1.442322977
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics       6.563927620
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics      4.540351421
## Grasslands-Cropland/Natural Vegetation Mosaics                        2.481223908
## Open Shrublands-Cropland/Natural Vegetation Mosaics                   7.088192621
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics           -1.753484274
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics               -1.778231890
## Savannas-Cropland/Natural Vegetation Mosaics                          1.929982367
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics         -1.623669127
## Water Bodies-Cropland/Natural Vegetation Mosaics                     -1.770054417
## Woody Savannas-Cropland/Natural Vegetation Mosaics                    2.902097478
## Evergreen Broadleaf Forests-Croplands                                8.006250597
## Evergreen Needleleaf Forests-Croplands                               5.982674398
## Grasslands-Croplands                                                 3.923546885
## Open Shrublands-Croplands                                            8.530515597
## Permanent Snow and Ice-Croplands                                    -0.311161297
## Permanent Wetlands-Croplands                                        -0.335908913
## Savannas-Croplands                                                   3.372305344
## Urban and Built-up Lands-Croplands                                  -0.181346150
## Water Bodies-Croplands                                              -0.327731440
## Woody Savannas-Croplands                                             4.344420455
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests            -2.023576199
## Grasslands-Evergreen Broadleaf Forests                              -4.082703711
## Open Shrublands-Evergreen Broadleaf Forests                          0.524265001
## Permanent Snow and Ice-Evergreen Broadleaf Forests                  -8.317411893
## Permanent Wetlands-Evergreen Broadleaf Forests                      -8.342159510
## Savannas-Evergreen Broadleaf Forests                                -4.633945252
## Urban and Built-up Lands-Evergreen Broadleaf Forests                -8.187596747
## Water Bodies-Evergreen Broadleaf Forests                            -8.333982037
## Woody Savannas-Evergreen Broadleaf Forests                          -3.661830142
## Grasslands-Evergreen Needleleaf Forests                             -2.059127513
## Open Shrublands-Evergreen Needleleaf Forests                         2.547841200
## Permanent Snow and Ice-Evergreen Needleleaf Forests                 -6.293835695
## Permanent Wetlands-Evergreen Needleleaf Forests                     -6.318583311
## Savannas-Evergreen Needleleaf Forests                               -2.610369054
## Urban and Built-up Lands-Evergreen Needleleaf Forests               -6.164020548
## Water Bodies-Evergreen Needleleaf Forests                           -6.310405838
## Woody Savannas-Evergreen Needleleaf Forests                         -1.638253943
## Open Shrublands-Grasslands                                           4.606968712
## Permanent Snow and Ice-Grasslands                                   -4.234708182
## Permanent Wetlands-Grasslands                                       -4.259455798
## Savannas-Grasslands                                                 -0.551241541
## Urban and Built-up Lands-Grasslands                                 -4.104893035
```

```
## Water Bodies-Grasslands                                          -4.251278325
## Woody Savannas-Grasslands                                         0.420873569
## Permanent Snow and Ice-Open Shrublands                           -8.841676894
## Permanent Wetlands-Open Shrublands                               -8.866424511
## Savannas-Open Shrublands                                         -5.158210253
## Urban and Built-up Lands-Open Shrublands                         -8.711861748
## Water Bodies-Open Shrublands                                     -8.858247037
## Woody Savannas-Open Shrublands                                   -4.186095143
## Permanent Wetlands-Permanent Snow and Ice                        -0.024747616
## Savannas-Permanent Snow and Ice                                   3.683466641
## Urban and Built-up Lands-Permanent Snow and Ice                   0.129815147
## Water Bodies-Permanent Snow and Ice                              -0.016570143
## Woody Savannas-Permanent Snow and Ice                             4.655581751
## Savannas-Permanent Wetlands                                       3.708214257
## Urban and Built-up Lands-Permanent Wetlands                       0.154562763
## Water Bodies-Permanent Wetlands                                   0.008177473
## Woody Savannas-Permanent Wetlands                                 4.680329368
## Urban and Built-up Lands-Savannas                                -3.553651495
## Water Bodies-Savannas                                            -3.700036784
## Woody Savannas-Savannas                                           0.972115110
## Water Bodies-Urban and Built-up Lands                            -0.146385290
## Woody Savannas-Urban and Built-up Lands                           4.525766605
## Woody Savannas-Water Bodies                                       4.672151895
##                                                                           lwr
## Closed Shrublands-Barren                                          4.63296061
## Cropland/Natural Vegetation Mosaics-Barren                      -12.44428656
## Croplands-Barren                                                 -3.65933286
## Evergreen Broadleaf Forests-Barren                               -6.59161816
## Evergreen Needleleaf Forests-Barren                               1.68321404
## Grasslands-Barren                                                 0.24745649
## Open Shrublands-Barren                                            3.20579043
## Permanent Snow and Ice-Barren                                   -19.69154320
## Permanent Wetlands-Barren                                       -16.79982145
## Savannas-Barren                                                  -1.07018302
## Urban and Built-up Lands-Barren                                 -12.92819242
## Water Bodies-Barren                                             -25.22995313
## Woody Savannas-Barren                                            -0.09806791
## Cropland/Natural Vegetation Mosaics-Closed Shrublands           -22.58995557
## Croplands-Closed Shrublands                                     -13.95453861
## Evergreen Broadleaf Forests-Closed Shrublands                   -16.73447510
## Evergreen Needleleaf Forests-Closed Shrublands                   -8.58376794
## Grasslands-Closed Shrublands                                    -10.04690099
## Open Shrublands-Closed Shrublands                                -7.02890552
## Permanent Snow and Ice-Closed Shrublands                        -29.82076687
## Permanent Wetlands-Closed Shrublands                            -26.93643390
## Savannas-Closed Shrublands                                      -11.33185469
## Urban and Built-up Lands-Closed Shrublands                      -23.07900591
## Water Bodies-Closed Shrublands                                  -35.34987222
## Woody Savannas-Closed Shrublands                                -10.35973958
## Croplands-Cropland/Natural Vegetation Mosaics                   -15.09623828
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics -13.21561956
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics -9.30963594
## Grasslands-Cropland/Natural Vegetation Mosaics                  -11.17747774
## Open Shrublands-Cropland/Natural Vegetation Mosaics              -7.13034985
```

```
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics       -25.33226950
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics           -22.99289990
## Savannas-Cropland/Natural Vegetation Mosaics                     -11.96746707
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics     -20.05605343
## Water Bodies-Cropland/Natural Vegetation Mosaics                 -30.10822746
## Woody Savannas-Cropland/Natural Vegetation Mosaics               -10.99535196
## Evergreen Broadleaf Forests-Croplands                            -6.38195573
## Evergreen Needleleaf Forests-Croplands                            3.22366287
## Grasslands-Croplands                                              2.39053507
## Open Shrublands-Croplands                                         4.29295207
## Permanent Snow and Ice-Croplands                                -19.59195763
## Permanent Wetlands-Croplands                                    -16.64088157
## Savannas-Croplands                                                0.38414611
## Urban and Built-up Lands-Croplands                              -12.65302549
## Water Bodies-Croplands                                          -25.20426750
## Woody Savannas-Croplands                                          1.35626122
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests        -16.59797938
## Grasslands-Evergreen Broadleaf Forests                          -18.47545219
## Open Shrublands-Evergreen Broadleaf Forests                     -14.40081551
## Permanent Snow and Ice-Evergreen Broadleaf Forests              -32.32887147
## Permanent Wetlands-Evergreen Broadleaf Forests                  -30.03670299
## Savannas-Evergreen Broadleaf Forests                            -19.25345890
## Urban and Built-up Lands-Evergreen Broadleaf Forests            -27.17032230
## Water Bodies-Evergreen Broadleaf Forests                        -37.03316547
## Woody Savannas-Evergreen Broadleaf Forests                      -18.28134379
## Grasslands-Evergreen Needleleaf Forests                          -4.84172920
## Open Shrublands-Evergreen Needleleaf Forests                     -2.28431066
## Permanent Snow and Ice-Evergreen Needleleaf Forests             -25.71397612
## Permanent Wetlands-Evergreen Needleleaf Forests                 -22.78809698
## Savannas-Evergreen Needleleaf Forests                            -6.39479009
## Urban and Built-up Lands-Evergreen Needleleaf Forests           -18.85005746
## Water Bodies-Evergreen Needleleaf Forests                       -31.29509701
## Woody Savannas-Evergreen Needleleaf Forests                      -5.42267498
## Open Shrublands-Grasslands                                        0.35400830
## Permanent Snow and Ice-Grasslands                               -23.51889431
## Permanent Wetlands-Grasslands                                   -20.56843679
## Savannas-Grasslands                                              -3.56119556
## Urban and Built-up Lands-Grasslands                             -16.58181226
## Water Bodies-Grasslands                                         -29.13044177
## Woody Savannas-Grasslands                                        -2.58908045
## Permanent Snow and Ice-Open Shrublands                          -28.52635759
## Permanent Wetlands-Open Shrublands                              -25.64705860
## Savannas-Open Shrublands                                        -10.12476264
## Urban and Built-up Lands-Open Shrublands                        -21.79927298
## Water Bodies-Open Shrublands                                    -34.04911032
## Woody Savannas-Open Shrublands                                   -9.15264753
## Permanent Wetlands-Permanent Snow and Ice                       -25.23152930
## Savannas-Permanent Snow and Ice                                 -15.77055107
## Urban and Built-up Lands-Permanent Snow and Ice                 -22.78464525
## Water Bodies-Permanent Snow and Ice                             -31.45495044
## Woody Savannas-Permanent Snow and Ice                           -14.79843596
## Savannas-Permanent Wetlands                                     -12.80123247
## Urban and Built-up Lands-Permanent Wetlands                     -20.31921367
## Water Bodies-Permanent Wetlands                                 -29.69829963
```

```
## Woody Savannas-Permanent Wetlands                                -11.82911736
## Urban and Built-up Lands-Savannas                                -16.29148819
## Water Bodies-Savannas                                            -28.71106923
## Woody Savannas-Savannas                                           -2.98247635
## Water Bodies-Urban and Built-up Lands                            -27.93425016
## Woody Savannas-Urban and Built-up Lands                           -8.21207009
## Woody Savannas-Water Bodies                                      -20.33888055
##                                                                          upr
## Closed Shrublands-Barren                                         15.5406517
## Cropland/Natural Vegetation Mosaics-Barren                       15.7946281
## Croplands-Barren                                                  4.1250285
## Evergreen Broadleaf Forests-Barren                               23.0698150
## Evergreen Needleleaf Forests-Barren                              10.7478304
## Grasslands-Barren                                                 8.0653329
## Open Shrublands-Barren                                           14.3209364
## Permanent Snow and Ice-Barren                                    19.5349162
## Permanent Wetlands-Barren                                        16.5936992
## Savannas-Barren                                                   8.2804893
## Urban and Built-up Lands-Barren                                  13.0311957
## Water Bodies-Barren                                              25.0401859
## Woody Savannas-Barren                                             9.2526044
## Cropland/Natural Vegetation Mosaics-Closed Shrublands             5.7666848
## Croplands-Closed Shrublands                                      -5.7533781
## Evergreen Broadleaf Forests-Closed Shrublands                    13.0390596
## Evergreen Needleleaf Forests-Closed Shrublands                    0.8412001
## Grasslands-Closed Shrublands                                     -1.8139219
## Open Shrublands-Closed Shrublands                                 4.3820200
## Permanent Snow and Ice-Closed Shrublands                          9.4905276
## Permanent Wetlands-Closed Shrublands                              6.5566994
## Savannas-Closed Shrublands                                       -1.6314513
## Urban and Built-up Lands-Closed Shrublands                        3.0083969
## Water Bodies-Closed Shrublands                                   14.9864927
## Woody Savannas-Closed Shrublands                                 -0.6593362
## Croplands-Cropland/Natural Vegetation Mosaics                    12.2115923
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics  26.3434748
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics 18.3903388
## Grasslands-Cropland/Natural Vegetation Mosaics                   16.1399256
## Open Shrublands-Cropland/Natural Vegetation Mosaics              21.3067351
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics       21.8253009
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics           19.4364361
## Savannas-Cropland/Natural Vegetation Mosaics                     15.8274318
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics     16.8087152
## Water Bodies-Cropland/Natural Vegetation Mosaics                 26.5681186
## Woody Savannas-Cropland/Natural Vegetation Mosaics               16.7995469
## Evergreen Broadleaf Forests-Croplands                            22.3944569
## Evergreen Needleleaf Forests-Croplands                            8.7416859
## Grasslands-Croplands                                              5.4565587
## Open Shrublands-Croplands                                        12.7680791
## Permanent Snow and Ice-Croplands                                 18.9696350
## Permanent Wetlands-Croplands                                     15.9690637
## Savannas-Croplands                                                6.3604646
## Urban and Built-up Lands-Croplands                               12.2903332
## Water Bodies-Croplands                                           24.5488046
## Woody Savannas-Croplands                                          7.3325797
```

```
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests         12.5508270
## Grasslands-Evergreen Broadleaf Forests                           10.3100448
## Open Shrublands-Evergreen Broadleaf Forests                      15.4493455
## Permanent Snow and Ice-Evergreen Broadleaf Forests               15.6940477
## Permanent Wetlands-Evergreen Broadleaf Forests                   13.3523840
## Savannas-Evergreen Broadleaf Forests                              9.9855684
## Urban and Built-up Lands-Evergreen Broadleaf Forests             10.7951288
## Water Bodies-Evergreen Broadleaf Forests                         20.3652014
## Woody Savannas-Evergreen Broadleaf Forests                       10.9576835
## Grasslands-Evergreen Needleleaf Forests                           0.7234742
## Open Shrublands-Evergreen Needleleaf Forests                      7.3799931
## Permanent Snow and Ice-Evergreen Needleleaf Forests              13.1263047
## Permanent Wetlands-Evergreen Needleleaf Forests                  10.1509304
## Savannas-Evergreen Needleleaf Forests                             1.1740520
## Urban and Built-up Lands-Evergreen Needleleaf Forests             6.5220164
## Water Bodies-Evergreen Needleleaf Forests                        18.6742853
## Woody Savannas-Evergreen Needleleaf Forests                       2.1461671
## Open Shrublands-Grasslands                                        8.8599291
## Permanent Snow and Ice-Grasslands                                15.0494780
## Permanent Wetlands-Grasslands                                    12.0495252
## Savannas-Grasslands                                               2.4587125
## Urban and Built-up Lands-Grasslands                               8.3720262
## Water Bodies-Grasslands                                          20.6278851
## Woody Savannas-Grasslands                                         3.4308276
## Permanent Snow and Ice-Open Shrublands                           10.8430038
## Permanent Wetlands-Open Shrublands                                7.9142096
## Savannas-Open Shrublands                                         -0.1916579
## Urban and Built-up Lands-Open Shrublands                          4.3755495
## Water Bodies-Open Shrublands                                     16.3326163
## Woody Savannas-Open Shrublands                                    0.7804572
## Permanent Wetlands-Permanent Snow and Ice                        25.1820341
## Savannas-Permanent Snow and Ice                                  23.1374844
## Urban and Built-up Lands-Permanent Snow and Ice                  23.0442755
## Water Bodies-Permanent Snow and Ice                              31.4218102
## Woody Savannas-Permanent Snow and Ice                            24.1095995
## Savannas-Permanent Wetlands                                      20.2176610
## Urban and Built-up Lands-Permanent Wetlands                      20.6283392
## Water Bodies-Permanent Wetlands                                  29.7146546
## Woody Savannas-Permanent Wetlands                                21.1897761
## Urban and Built-up Lands-Savannas                                 9.1841852
## Water Bodies-Savannas                                            21.3109957
## Woody Savannas-Savannas                                           4.9267066
## Water Bodies-Urban and Built-up Lands                            27.6414796
## Woody Savannas-Urban and Built-up Lands                          17.2636033
## Woody Savannas-Water Bodies                                      29.6831843
##                                                                      p adj
## Closed Shrublands-Barren                                         0.0000001
## Cropland/Natural Vegetation Mosaics-Barren                       1.0000000
## Croplands-Barren                                                 1.0000000
## Evergreen Broadleaf Forests-Barren                               0.8458049
## Evergreen Needleleaf Forests-Barren                              0.0003683
## Grasslands-Barren                                                0.0249025
## Open Shrublands-Barren                                           0.0000114
## Permanent Snow and Ice-Barren                                    1.0000000
```

```
## Permanent Wetlands-Barren                                          1.0000000
## Savannas-Barren                                                    0.3486088
## Urban and Built-up Lands-Barren                                    1.0000000
## Water Bodies-Barren                                                1.0000000
## Woody Savannas-Barren                                              0.0621011
## Cropland/Natural Vegetation Mosaics-Closed Shrublands              0.7738116
## Croplands-Closed Shrublands                                        0.0000000
## Evergreen Broadleaf Forests-Closed Shrublands                      1.0000000
## Evergreen Needleleaf Forests-Closed Shrublands                     0.2473194
## Grasslands-Closed Shrublands                                       0.0001206
## Open Shrublands-Closed Shrublands                                  0.9999593
## Permanent Snow and Ice-Closed Shrublands                           0.9034304
## Permanent Wetlands-Closed Shrublands                               0.7410664
## Savannas-Closed Shrublands                                         0.0006332
## Urban and Built-up Lands-Closed Shrublands                         0.3524449
## Water Bodies-Closed Shrublands                                     0.9863675
## Woody Savannas-Closed Shrublands                                   0.0103776
## Croplands-Cropland/Natural Vegetation Mosaics                      1.0000000
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics    0.9979582
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics   0.9981973
## Grasslands-Cropland/Natural Vegetation Mosaics                     0.9999977
## Open Shrublands-Cropland/Natural Vegetation Mosaics                0.9254938
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics         1.0000000
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics             1.0000000
## Savannas-Cropland/Natural Vegetation Mosaics                       0.9999999
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics       1.0000000
## Water Bodies-Cropland/Natural Vegetation Mosaics                   1.0000000
## Woody Savannas-Cropland/Natural Vegetation Mosaics                 0.9999881
## Evergreen Broadleaf Forests-Croplands                             0.8442536
## Evergreen Needleleaf Forests-Croplands                            0.0000000
## Grasslands-Croplands                                              0.0000000
## Open Shrublands-Croplands                                         0.0000000
## Permanent Snow and Ice-Croplands                                  1.0000000
## Permanent Wetlands-Croplands                                      1.0000000
## Savannas-Croplands                                                0.0113836
## Urban and Built-up Lands-Croplands                                1.0000000
## Water Bodies-Croplands                                            1.0000000
## Woody Savannas-Croplands                                          0.0000968
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests          0.9999999
## Grasslands-Evergreen Broadleaf Forests                            0.9996091
## Open Shrublands-Evergreen Broadleaf Forests                       1.0000000
## Permanent Snow and Ice-Evergreen Broadleaf Forests                0.9968588
## Permanent Wetlands-Evergreen Broadleaf Forests                    0.9914200
## Savannas-Evergreen Broadleaf Forests                              0.9987260
## Urban and Built-up Lands-Evergreen Broadleaf Forests              0.9761721
## Water Bodies-Evergreen Broadleaf Forests                          0.9994961
## Woody Savannas-Evergreen Broadleaf Forests                        0.9999019
## Grasslands-Evergreen Needleleaf Forests                           0.4199921
## Open Shrublands-Evergreen Needleleaf Forests                      0.8898534
## Permanent Snow and Ice-Evergreen Needleleaf Forests               0.9983972
## Permanent Wetlands-Evergreen Needleleaf Forests                   0.9915999
## Savannas-Evergreen Needleleaf Forests                             0.5440586
## Urban and Built-up Lands-Evergreen Needleleaf Forests             0.9383073
## Water Bodies-Evergreen Needleleaf Forests                         0.9998922
```

```
## Woody Savannas-Evergreen Needleleaf Forests                        0.9754256
## Open Shrublands-Grasslands                                         0.0197388
## Permanent Snow and Ice-Grasslands                                  0.9999784
## Permanent Wetlands-Grasslands                                      0.9998429
## Savannas-Grasslands                                                0.9999975
## Urban and Built-up Lands-Grasslands                                0.9981302
## Water Bodies-Grasslands                                            0.9999989
## Woody Savannas-Grasslands                                          0.9999999
## Permanent Snow and Ice-Open Shrublands                             0.9666227
## Permanent Wetlands-Open Shrublands                                 0.8883011
## Savannas-Open Shrublands                                           0.0329382
## Urban and Built-up Lands-Open Shrublands                           0.6046739
## Water Bodies-Open Shrublands                                       0.9963549
## Woody Savannas-Open Shrublands                                     0.2106297
## Permanent Wetlands-Permanent Snow and Ice                          1.0000000
## Savannas-Permanent Snow and Ice                                    0.9999963
## Urban and Built-up Lands-Permanent Snow and Ice                    1.0000000
## Water Bodies-Permanent Snow and Ice                                1.0000000
## Woody Savannas-Permanent Snow and Ice                              0.9999416
## Savannas-Permanent Wetlands                                        0.9999719
## Urban and Built-up Lands-Permanent Wetlands                        1.0000000
## Water Bodies-Permanent Wetlands                                    1.0000000
## Woody Savannas-Permanent Wetlands                                  0.9996116
## Urban and Built-up Lands-Savannas                                  0.9996740
## Water Bodies-Savannas                                              0.9999998
## Woody Savannas-Savannas                                            0.9999207
## Water Bodies-Urban and Built-up Lands                              1.0000000
## Woody Savannas-Urban and Built-up Lands                            0.9959655
## Woody Savannas-Water Bodies                                        0.9999968
```

```
TukeyHSD(min_model)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = mn_grw_km2 ~ lc_name, data = fire_events)
##
## $lc_name
##                                                                      diff
## Closed Shrublands-Barren                                     6.570959e-02
## Cropland/Natural Vegetation Mosaics-Barren                  -3.252404e-02
## Croplands-Barren                                             8.608159e-03
## Evergreen Broadleaf Forests-Barren                           3.902885e-02
## Evergreen Needleleaf Forests-Barren                          8.901317e-03
## Grasslands-Barren                                            1.052656e-01
## Open Shrublands-Barren                                       2.055519e-01
## Permanent Snow and Ice-Barren                               -3.252404e-02
## Permanent Wetlands-Barren                                    2.880701e-02
## Savannas-Barren                                              7.163523e-02
## Urban and Built-up Lands-Barren                              1.821346e-01
## Water Bodies-Barren                                          3.902885e-02
## Woody Savannas-Barren                                        4.989764e-02
## Cropland/Natural Vegetation Mosaics-Closed Shrublands       -9.823363e-02
## Croplands-Closed Shrublands                                 -5.710143e-02
```

```
## Evergreen Broadleaf Forests-Closed Shrublands                    -2.668074e-02
## Evergreen Needleleaf Forests-Closed Shrublands                   -5.680827e-02
## Grasslands-Closed Shrublands                                      3.955597e-02
## Open Shrublands-Closed Shrublands                                 1.398424e-01
## Permanent Snow and Ice-Closed Shrublands                         -9.823363e-02
## Permanent Wetlands-Closed Shrublands                             -3.690258e-02
## Savannas-Closed Shrublands                                        5.925642e-03
## Urban and Built-up Lands-Closed Shrublands                        1.164250e-01
## Water Bodies-Closed Shrublands                                   -2.668074e-02
## Woody Savannas-Closed Shrublands                                 -1.581195e-02
## Croplands-Cropland/Natural Vegetation Mosaics                     4.113220e-02
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics   7.155289e-02
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics  4.142536e-02
## Grasslands-Cropland/Natural Vegetation Mosaics                    1.377896e-01
## Open Shrublands-Cropland/Natural Vegetation Mosaics               2.380760e-01
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics       -8.326673e-17
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics            6.133105e-02
## Savannas-Cropland/Natural Vegetation Mosaics                      1.041593e-01
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics      2.146587e-01
## Water Bodies-Cropland/Natural Vegetation Mosaics                  7.155289e-02
## Woody Savannas-Cropland/Natural Vegetation Mosaics                8.242168e-02
## Evergreen Broadleaf Forests-Croplands                            3.042069e-02
## Evergreen Needleleaf Forests-Croplands                           2.931571e-04
## Grasslands-Croplands                                             9.665740e-02
## Open Shrublands-Croplands                                        1.969438e-01
## Permanent Snow and Ice-Croplands                                -4.113220e-02
## Permanent Wetlands-Croplands                                     2.019885e-02
## Savannas-Croplands                                               6.302707e-02
## Urban and Built-up Lands-Croplands                               1.735265e-01
## Water Bodies-Croplands                                           3.042069e-02
## Woody Savannas-Croplands                                         4.128948e-02
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests        -3.012753e-02
## Grasslands-Evergreen Broadleaf Forests                           6.623671e-02
## Open Shrublands-Evergreen Broadleaf Forests                      1.665231e-01
## Permanent Snow and Ice-Evergreen Broadleaf Forests              -7.155289e-02
## Permanent Wetlands-Evergreen Broadleaf Forests                  -1.022184e-02
## Savannas-Evergreen Broadleaf Forests                             3.260638e-02
## Urban and Built-up Lands-Evergreen Broadleaf Forests             1.431058e-01
## Water Bodies-Evergreen Broadleaf Forests                         1.148526e-13
## Woody Savannas-Evergreen Broadleaf Forests                       1.086879e-02
## Grasslands-Evergreen Needleleaf Forests                          9.636425e-02
## Open Shrublands-Evergreen Needleleaf Forests                     1.966506e-01
## Permanent Snow and Ice-Evergreen Needleleaf Forests             -4.142536e-02
## Permanent Wetlands-Evergreen Needleleaf Forests                  1.990569e-02
## Savannas-Evergreen Needleleaf Forests                            6.273391e-02
## Urban and Built-up Lands-Evergreen Needleleaf Forests            1.732333e-01
## Water Bodies-Evergreen Needleleaf Forests                        3.012753e-02
## Woody Savannas-Evergreen Needleleaf Forests                      4.099633e-02
## Open Shrublands-Grasslands                                       1.002864e-01
## Permanent Snow and Ice-Grasslands                               -1.377896e-01
## Permanent Wetlands-Grasslands                                   -7.645855e-02
## Savannas-Grasslands                                             -3.363033e-02
## Urban and Built-up Lands-Grasslands                              7.686907e-02
## Water Bodies-Grasslands                                         -6.623671e-02
```

```
## Woody Savannas-Grasslands                                          -5.536792e-02
## Permanent Snow and Ice-Open Shrublands                             -2.380760e-01
## Permanent Wetlands-Open Shrublands                                 -1.767449e-01
## Savannas-Open Shrublands                                           -1.339167e-01
## Urban and Built-up Lands-Open Shrublands                           -2.341731e-02
## Water Bodies-Open Shrublands                                       -1.665231e-01
## Woody Savannas-Open Shrublands                                     -1.556543e-01
## Permanent Wetlands-Permanent Snow and Ice                           6.133105e-02
## Savannas-Permanent Snow and Ice                                     1.041593e-01
## Urban and Built-up Lands-Permanent Snow and Ice                     2.146587e-01
## Water Bodies-Permanent Snow and Ice                                 7.155289e-02
## Woody Savannas-Permanent Snow and Ice                               8.242168e-02
## Savannas-Permanent Wetlands                                         4.282822e-02
## Urban and Built-up Lands-Permanent Wetlands                         1.533276e-01
## Water Bodies-Permanent Wetlands                                     1.022184e-02
## Woody Savannas-Permanent Wetlands                                   2.109064e-02
## Urban and Built-up Lands-Savannas                                   1.104994e-01
## Water Bodies-Savannas                                              -3.260638e-02
## Woody Savannas-Savannas                                            -2.173759e-02
## Water Bodies-Urban and Built-up Lands                              -1.431058e-01
## Woody Savannas-Urban and Built-up Lands                            -1.322370e-01
## Woody Savannas-Water Bodies                                         1.086879e-02
##                                                                             lwr
## Closed Shrublands-Barren                                          -0.07882574
## Cropland/Natural Vegetation Mosaics-Barren                        -0.40671148
## Croplands-Barren                                                  -0.09454064
## Evergreen Broadleaf Forests-Barren                                -0.35400805
## Evergreen Needleleaf Forests-Barren                               -0.11121185
## Grasslands-Barren                                                  0.00167266
## Open Shrublands-Barren                                             0.05826767
## Permanent Snow and Ice-Barren                                     -0.55230493
## Permanent Wetlands-Barren                                         -0.41368294
## Savannas-Barren                                                   -0.05226840
## Urban and Built-up Lands-Barren                                   -0.16184731
## Water Bodies-Barren                                               -0.62708933
## Woody Savannas-Barren                                             -0.07400599
## Cropland/Natural Vegetation Mosaics-Closed Shrublands             -0.47398102
## Croplands-Closed Shrublands                                       -0.16577314
## Evergreen Broadleaf Forests-Closed Shrublands                     -0.42120307
## Evergreen Needleleaf Forests-Closed Shrublands                    -0.18169638
## Grasslands-Closed Shrublands                                      -0.06953736
## Open Shrublands-Closed Shrublands                                 -0.01136123
## Permanent Snow and Ice-Closed Shrublands                          -0.61913865
## Permanent Wetlands-Closed Shrublands                              -0.48071247
## Savannas-Closed Shrublands                                        -0.12261220
## Urban and Built-up Lands-Closed Shrublands                        -0.22925320
## Water Bodies-Closed Shrublands                                    -0.69367646
## Woody Savannas-Closed Shrublands                                  -0.14434978
## Croplands-Cropland/Natural Vegetation Mosaics                     -0.32071765
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics   -0.45263567
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics  -0.32562071
## Grasslands-Cropland/Natural Vegetation Mosaics                    -0.22418709
## Open Shrublands-Cropland/Natural Vegetation Mosaics               -0.13873736
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics        -0.62487424
```

```
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics           -0.50089043
## Savannas-Cropland/Natural Vegetation Mosaics                     -0.26414461
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics     -0.27382799
## Water Bodies-Cropland/Natural Vegetation Mosaics                 -0.67945248
## Woody Savannas-Cropland/Natural Vegetation Mosaics               -0.28588220
## Evergreen Broadleaf Forests-Croplands                           -0.35088901
## Evergreen Needleleaf Forests-Croplands                          -0.07282491
## Grasslands-Croplands                                             0.05603022
## Open Shrublands-Croplands                                        0.08464180
## Permanent Snow and Ice-Croplands                                -0.55210309
## Permanent Wetlands-Croplands                                    -0.41190812
## Savannas-Croplands                                              -0.01616377
## Urban and Built-up Lands-Croplands                              -0.15699230
## Water Bodies-Croplands                                          -0.62884595
## Woody Savannas-Croplands                                        -0.03790135
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests        -0.41637174
## Grasslands-Evergreen Broadleaf Forests                          -0.31519336
## Open Shrublands-Evergreen Broadleaf Forests                     -0.22901460
## Permanent Snow and Ice-Evergreen Broadleaf Forests              -0.70789367
## Permanent Wetlands-Evergreen Broadleaf Forests                  -0.58516076
## Savannas-Evergreen Broadleaf Forests                            -0.35483332
## Urban and Built-up Lands-Evergreen Broadleaf Forests            -0.35996577
## Water Bodies-Evergreen Broadleaf Forests                        -0.76057270
## Woody Savannas-Evergreen Broadleaf Forests                      -0.37657091
## Grasslands-Evergreen Needleleaf Forests                          0.02262100
## Open Shrublands-Evergreen Needleleaf Forests                     0.06859114
## Permanent Snow and Ice-Evergreen Needleleaf Forests             -0.55608908
## Permanent Wetlands-Evergreen Needleleaf Forests                 -0.41656187
## Savannas-Evergreen Needleleaf Forests                           -0.03755909
## Urban and Built-up Lands-Evergreen Needleleaf Forests           -0.16296626
## Water Bodies-Evergreen Needleleaf Forests                       -0.63200538
## Woody Savannas-Evergreen Needleleaf Forests                     -0.05929668
## Open Shrublands-Grasslands                                      -0.01242364
## Permanent Snow and Ice-Grasslands                               -0.64885033
## Permanent Wetlands-Grasslands                                   -0.50867175
## Savannas-Grasslands                                             -0.11339876
## Urban and Built-up Lands-Grasslands                             -0.25378857
## Water Bodies-Grasslands                                         -0.72557298
## Woody Savannas-Grasslands                                       -0.13513635
## Permanent Snow and Ice-Open Shrublands                          -0.75975043
## Permanent Wetlands-Open Shrublands                              -0.62145766
## Savannas-Open Shrublands                                        -0.26553802
## Urban and Built-up Lands-Open Shrublands                        -0.37025393
## Water Bodies-Open Shrublands                                    -0.83411989
## Woody Savannas-Open Shrublands                                  -0.28727561
## Permanent Wetlands-Permanent Snow and Ice                       -0.60668761
## Savannas-Permanent Snow and Ice                                 -0.41140225
## Urban and Built-up Lands-Permanent Snow and Ice                 -0.39260993
## Water Bodies-Permanent Snow and Ice                             -0.76161276
## Woody Savannas-Permanent Snow and Ice                           -0.43313984
## Savannas-Permanent Wetlands                                     -0.39469762
## Urban and Built-up Lands-Permanent Wetlands                     -0.38925908
## Water Bodies-Permanent Wetlands                                 -0.77704570
## Woody Savannas-Permanent Wetlands                               -0.41643521
```

```
## Urban and Built-up Lands-Savannas                                -0.22707295
## Water Bodies-Savannas                                            -0.69543738
## Woody Savannas-Savannas                                          -0.12654037
## Water Bodies-Urban and Built-up Lands                            -0.87952713
## Woody Savannas-Urban and Built-up Lands                          -0.46980934
## Woody Savannas-Water Bodies                                      -0.65196221
##                                                                         upr
## Closed Shrublands-Barren                                          0.21024492
## Cropland/Natural Vegetation Mosaics-Barren                        0.34166339
## Croplands-Barren                                                  0.11175696
## Evergreen Broadleaf Forests-Barren                                0.43206575
## Evergreen Needleleaf Forests-Barren                               0.12901449
## Grasslands-Barren                                                 0.20885846
## Open Shrublands-Barren                                            0.35283621
## Permanent Snow and Ice-Barren                                     0.48725685
## Permanent Wetlands-Barren                                         0.47129695
## Savannas-Barren                                                   0.19553886
## Urban and Built-up Lands-Barren                                   0.52611658
## Water Bodies-Barren                                               0.70514703
## Woody Savannas-Barren                                             0.17380128
## Cropland/Natural Vegetation Mosaics-Closed Shrublands             0.27751376
## Croplands-Closed Shrublands                                       0.05157028
## Evergreen Broadleaf Forests-Closed Shrublands                     0.36784160
## Evergreen Needleleaf Forests-Closed Shrublands                    0.06807983
## Grasslands-Closed Shrublands                                      0.14864931
## Open Shrublands-Closed Shrublands                                 0.29104593
## Permanent Snow and Ice-Closed Shrublands                          0.42267139
## Permanent Wetlands-Closed Shrublands                              0.40690731
## Savannas-Closed Shrublands                                        0.13446348
## Urban and Built-up Lands-Closed Shrublands                        0.46210328
## Water Bodies-Closed Shrublands                                    0.64031498
## Woody Savannas-Closed Shrublands                                  0.11272589
## Croplands-Cropland/Natural Vegetation Mosaics                     0.40298205
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics   0.59574145
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics  0.40847142
## Grasslands-Cropland/Natural Vegetation Mosaics                    0.49976630
## Open Shrublands-Cropland/Natural Vegetation Mosaics               0.61488933
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics        0.62487424
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics            0.62355253
## Savannas-Cropland/Natural Vegetation Mosaics                      0.47246315
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics      0.70314534
## Water Bodies-Cropland/Natural Vegetation Mosaics                  0.82255826
## Woody Savannas-Cropland/Natural Vegetation Mosaics                0.45072557
## Evergreen Broadleaf Forests-Croplands                            0.41173039
## Evergreen Needleleaf Forests-Croplands                           0.07341123
## Grasslands-Croplands                                             0.13728458
## Open Shrublands-Croplands                                        0.30924576
## Permanent Snow and Ice-Croplands                                 0.46983869
## Permanent Wetlands-Croplands                                     0.45230582
## Savannas-Croplands                                               0.14221791
## Urban and Built-up Lands-Croplands                               0.50404524
## Water Bodies-Croplands                                           0.68968733
## Woody Savannas-Croplands                                         0.12048032
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests         0.35611667
```

```
## Grasslands-Evergreen Broadleaf Forests                          0.44766678
## Open Shrublands-Evergreen Broadleaf Forests                      0.56206078
## Permanent Snow and Ice-Evergreen Broadleaf Forests               0.56478789
## Permanent Wetlands-Evergreen Broadleaf Forests                   0.56471708
## Savannas-Evergreen Broadleaf Forests                             0.42004608
## Urban and Built-up Lands-Evergreen Broadleaf Forests             0.64617734
## Water Bodies-Evergreen Broadleaf Forests                         0.76057270
## Woody Savannas-Evergreen Broadleaf Forests                       0.39830849
## Grasslands-Evergreen Needleleaf Forests                          0.17010749
## Open Shrublands-Evergreen Needleleaf Forests                     0.32471012
## Permanent Snow and Ice-Evergreen Needleleaf Forests              0.47323837
## Permanent Wetlands-Evergreen Needleleaf Forests                  0.45637325
## Savannas-Evergreen Needleleaf Forests                            0.16302692
## Urban and Built-up Lands-Evergreen Needleleaf Forests            0.50943289
## Water Bodies-Evergreen Needleleaf Forests                        0.69226045
## Woody Savannas-Evergreen Needleleaf Forests                      0.14128933
## Open Shrublands-Grasslands                                       0.21299640
## Permanent Snow and Ice-Grasslands                                0.37327112
## Permanent Wetlands-Grasslands                                    0.35575464
## Savannas-Grasslands                                              0.04613810
## Urban and Built-up Lands-Grasslands                              0.40752671
## Water Bodies-Grasslands                                          0.59309956
## Woody Savannas-Grasslands                                        0.02440051
## Permanent Snow and Ice-Open Shrublands                           0.28359847
## Permanent Wetlands-Open Shrublands                               0.26796780
## Savannas-Open Shrublands                                        -0.00229540
## Urban and Built-up Lands-Open Shrublands                         0.32341931
## Water Bodies-Open Shrublands                                     0.50107371
## Woody Savannas-Open Shrublands                                  -0.02403299
## Permanent Wetlands-Permanent Snow and Ice                        0.72934971
## Savannas-Permanent Snow and Ice                                  0.61972080
## Urban and Built-up Lands-Permanent Snow and Ice                  0.82192728
## Water Bodies-Permanent Snow and Ice                              0.90471854
## Woody Savannas-Permanent Snow and Ice                            0.59798321
## Savannas-Permanent Wetlands                                      0.48035407
## Urban and Built-up Lands-Permanent Wetlands                      0.69591433
## Water Bodies-Permanent Wetlands                                  0.79748938
## Woody Savannas-Permanent Wetlands                                0.45861648
## Urban and Built-up Lands-Savannas                                0.44807175
## Water Bodies-Savannas                                            0.63022462
## Woody Savannas-Savannas                                          0.08306520
## Water Bodies-Urban and Built-up Lands                            0.59331557
## Woody Savannas-Urban and Built-up Lands                          0.20533536
## Woody Savannas-Water Bodies                                      0.67369980
##                                                                      p adj
## Closed Shrublands-Barren                                         0.9631905
## Cropland/Natural Vegetation Mosaics-Barren                       1.0000000
## Croplands-Barren                                                 1.0000000
## Evergreen Broadleaf Forests-Barren                               1.0000000
## Evergreen Needleleaf Forests-Barren                              1.0000000
## Grasslands-Barren                                                0.0421099
## Open Shrublands-Barren                                           0.0002509
## Permanent Snow and Ice-Barren                                    1.0000000
## Permanent Wetlands-Barren                                        1.0000000
```

```
## Savannas-Barren                                                    0.8043612
## Urban and Built-up Lands-Barren                                    0.8867173
## Water Bodies-Barren                                                1.0000000
## Woody Savannas-Barren                                              0.9869113
## Cropland/Natural Vegetation Mosaics-Closed Shrublands              0.9998411
## Croplands-Closed Shrublands                                        0.8923820
## Evergreen Broadleaf Forests-Closed Shrublands                      1.0000000
## Evergreen Needleleaf Forests-Closed Shrublands                     0.9630294
## Grasslands-Closed Shrublands                                       0.9950848
## Open Shrublands-Closed Shrublands                                  0.1050166
## Permanent Snow and Ice-Closed Shrublands                           0.9999965
## Permanent Wetlands-Closed Shrublands                               1.0000000
## Savannas-Closed Shrublands                                         1.0000000
## Urban and Built-up Lands-Closed Shrublands                         0.9976283
## Water Bodies-Closed Shrublands                                     1.0000000
## Woody Savannas-Closed Shrublands                                   1.0000000
## Croplands-Cropland/Natural Vegetation Mosaics                      1.0000000
## Evergreen Broadleaf Forests-Cropland/Natural Vegetation Mosaics    0.9999999
## Evergreen Needleleaf Forests-Cropland/Natural Vegetation Mosaics   1.0000000
## Grasslands-Cropland/Natural Vegetation Mosaics                     0.9921925
## Open Shrublands-Cropland/Natural Vegetation Mosaics                0.6874511
## Permanent Snow and Ice-Cropland/Natural Vegetation Mosaics         1.0000000
## Permanent Wetlands-Cropland/Natural Vegetation Mosaics             1.0000000
## Savannas-Cropland/Natural Vegetation Mosaics                       0.9996218
## Urban and Built-up Lands-Cropland/Natural Vegetation Mosaics       0.9721322
## Water Bodies-Cropland/Natural Vegetation Mosaics                   1.0000000
## Woody Savannas-Cropland/Natural Vegetation Mosaics                 0.9999731
## Evergreen Broadleaf Forests-Croplands                             1.0000000
## Evergreen Needleleaf Forests-Croplands                            1.0000000
## Grasslands-Croplands                                              0.0000000
## Open Shrublands-Croplands                                         0.0000004
## Permanent Snow and Ice-Croplands                                  1.0000000
## Permanent Wetlands-Croplands                                      1.0000000
## Savannas-Croplands                                                0.2963151
## Urban and Built-up Lands-Croplands                                0.8929830
## Water Bodies-Croplands                                            1.0000000
## Woody Savannas-Croplands                                          0.8978856
## Evergreen Needleleaf Forests-Evergreen Broadleaf Forests          1.0000000
## Grasslands-Evergreen Broadleaf Forests                            0.9999987
## Open Shrublands-Evergreen Broadleaf Forests                       0.9806335
## Permanent Snow and Ice-Evergreen Broadleaf Forests                1.0000000
## Permanent Wetlands-Evergreen Broadleaf Forests                    1.0000000
## Savannas-Evergreen Broadleaf Forests                              1.0000000
## Urban and Built-up Lands-Evergreen Broadleaf Forests              0.9995970
## Water Bodies-Evergreen Broadleaf Forests                          1.0000000
## Woody Savannas-Evergreen Broadleaf Forests                        1.0000000
## Grasslands-Evergreen Needleleaf Forests                           0.0009895
## Open Shrublands-Evergreen Needleleaf Forests                      0.0000238
## Permanent Snow and Ice-Evergreen Needleleaf Forests               1.0000000
## Permanent Wetlands-Evergreen Needleleaf Forests                   1.0000000
## Savannas-Evergreen Needleleaf Forests                             0.7022894
## Urban and Built-up Lands-Evergreen Needleleaf Forests             0.9058433
## Water Bodies-Evergreen Needleleaf Forests                         1.0000000
## Woody Savannas-Evergreen Needleleaf Forests                       0.9850468
```

```
## Open Shrublands-Grasslands                                    0.1435988
## Permanent Snow and Ice-Grasslands                             0.9997762
## Permanent Wetlands-Grasslands                                 0.9999984
## Savannas-Grasslands                                           0.9803953
## Urban and Built-up Lands-Grasslands                           0.9999582
## Water Bodies-Grasslands                                       1.0000000
## Woody Savannas-Grasslands                                     0.5331237
## Permanent Snow and Ice-Open Shrublands                        0.9620418
## Permanent Wetlands-Open Shrublands                            0.9883805
## Savannas-Open Shrublands                                      0.0415273
## Urban and Built-up Lands-Open Shrublands                      1.0000000
## Water Bodies-Open Shrublands                                  0.9999064
## Woody Savannas-Open Shrublands                                0.0056811
## Permanent Wetlands-Permanent Snow and Ice                     1.0000000
## Savannas-Permanent Snow and Ice                               0.9999919
## Urban and Built-up Lands-Permanent Snow and Ice               0.9961634
## Water Bodies-Permanent Snow and Ice                           1.0000000
## Woody Savannas-Permanent Snow and Ice                         0.9999995
## Savannas-Permanent Wetlands                                   1.0000000
## Urban and Built-up Lands-Permanent Wetlands                   0.9996250
## Water Bodies-Permanent Wetlands                               1.0000000
## Woody Savannas-Permanent Wetlands                             1.0000000
## Urban and Built-up Lands-Savannas                             0.9982246
## Water Bodies-Savannas                                         1.0000000
## Woody Savannas-Savannas                                       0.9999890
## Water Bodies-Urban and Built-up Lands                         0.9999949
## Woody Savannas-Urban and Built-up Lands                       0.9898167
## Woody Savannas-Water Bodies                                   1.0000000
```

```r
#Sorting the Tukey tests so they are more readable
mean_sig <- as.data.frame(TukeyHSD(mean_model)$lc_name) |>
  filter(`p adj` < 0.05) |>
  arrange(`p adj`)

min_sig <- as.data.frame(TukeyHSD(max_model)$lc_name) |>
  filter(`p adj` < 0.05) |>
  arrange(`p adj`)

max_sig <- as.data.frame(TukeyHSD(min_model)$lc_name) |>
  filter(`p adj` < 0.05) |>
  arrange(`p adj`)

tukey_summary <- data.frame(Metric = c("Maximum", "Mean", "Minimum"),
  Significant_comparisons = c(nrow(max_sig), nrow(mean_sig), nrow(min_sig)))
tukey_summary
```

```
##    Metric Significant_comparisons
## 1 Maximum                       8
## 2    Mean                      15
## 3 Minimum                      13
```

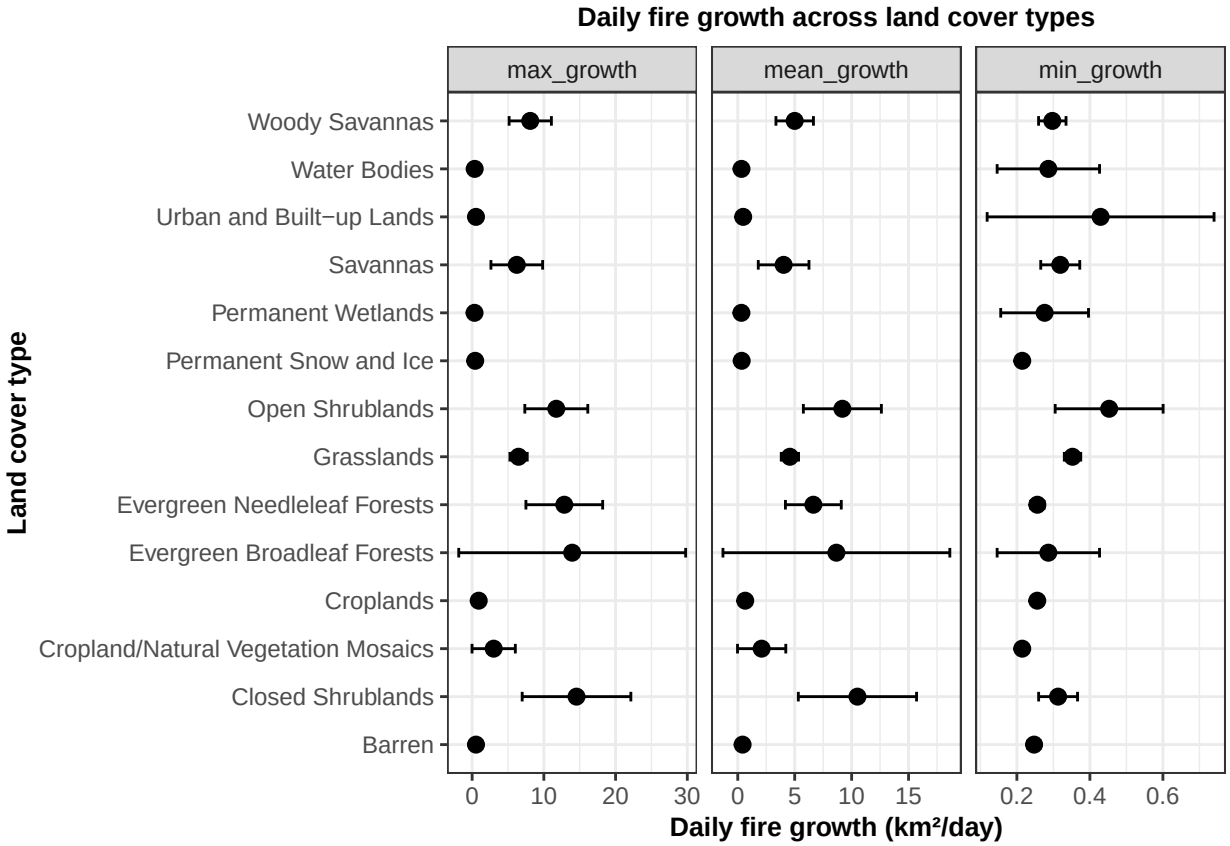
#The most strongly recurring pattern is open shrubland's significantly high fire growth as compared to other land cover types.They have a significantly higher mean daily growth than Croplands, Barren, and

Grasslands, a higher maximum daily growth than Croplands, Evergreen Needleleaf Forests, and Barren, and higher minimums than Croplands, Evergreen Needleleaf Forests, and Barren. Closed shrublands and evergreen needleleef forests also have relatively high mean and maximum daily growth levels, but not minimum indicating that they might have relatively high growth during active periods but not throughout event duration.

```
#Load tidyr
library(tidyr)

#make dataframe with mean, sd for each of the three categories and pivot it
growth_summary <- fire_events |>
  group_by(lc_name) |>
  summarize(
     max_growth = mean(mx_grw_km2, na.rm = TRUE),
    mean_growth = mean(mu_grw_km2, na.rm = TRUE),
    min_growth = mean(mn_grw_km2, na.rm = TRUE),
    max_se = sd(mx_grw_km2, na.rm = TRUE) / sqrt(sum(!is.na(mx_grw_km2))),
    mean_se = sd(mu_grw_km2, na.rm = TRUE) / sqrt(sum(!is.na(mu_grw_km2))),
    min_se = sd(mn_grw_km2, na.rm = TRUE) / sqrt(sum(!is.na(mn_grw_km2))),
    n = n(),
    .groups = "drop"
  )
se_long <- growth_summary |>
  pivot_longer(cols = c(max_se, mean_se, min_se),names_to = "metric",
    values_to = "se")|>
  mutate(metric = recode(metric,max_se = "max_growth",mean_se = "mean_growth",
      min_se = "min_growth"))
growth_long <- growth_summary |>
  pivot_longer(cols = c(max_growth, mean_growth, min_growth),names_to = "metric",
    values_to = "growth")
growth_long <- growth_long |>
  left_join(se_long, by = c("lc_name", "metric"))

#Make graphic plotting max/mean/min fire growth rates across land cover types
ggplot(growth_long, aes(x = growth, y = lc_name))+
  geom_point(size = 2.5) +
  geom_errorbar(aes(xmin = growth - 1.96 * se, xmax = growth + 1.96 * se), width = 0.2) +
  facet_wrap(~metric, scales = "free_x") +
  labs(x = "Daily fire growth (km²/day)",y = "Land cover type",
       title = "Daily fire growth across land cover types")
```



#This graphic plots daily fire growth on the X axis and land cover type per event on the X axis. As mentioned before, open shrubland's relatively high growth levels jumps out, especially given its significance after running the anova and tukey tests. Closed shrubland and evergreen needleleaf forests also have relatively high means. Some land cover types not mentioned earlier, like evergreen broadleaf forests and urban/built up lands also visually jump out as having possibly higher growth rates, but they were not found to be statistically significant after the Tukey tests, which is visible given the extremely high spread for both units.That represents one limitation of this figure - we felt it was important to include results for all of the land cover types so that the reader could parse all the data, but not all of them were statistically significant. The primary takeaway from this figure and the associated tables up above is to possibly take a deeper dive into open shrubland fire growth specifically, perhaps running some of the above tests filtered to examine only open shrubland. Examining needleleaf forests might provide useful results as well given their prevalence in California. "'